

GE Healthcare

LightSpeed 4.X

Pre-Installation Manual

This manual applies only to LightSpeed 4.X systems shipped in 2006 or later by GoldSeal.

Operating Documentation.



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Revision: 01

LEGAL NOTES

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fánleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkāpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkāpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkāpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumā apkāpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kılavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. Aug. 6, 2007

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
01	3/31/08	Initial release of document meeting all current and regulatory conditions for systems shipped in 2006 or later by GoldSeal.

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Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GE Healthcare's global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0

Conventions Used For General Information

2.1 Document Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout

Headers and footers in this publication are designed to allow you to quickly identify your location. The document’s part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

Section 3.0

Computer Screen Output and Input

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It’s output is fixed from the sense that it does not vary from application to application.

Example: *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example:*
<variable_ouput>

Example: This paragraph denotes fixed input. It’s typed input that will not vary from application to application.

Example: *This paragraph denotes computer input that can vary from application to application. Variable input will normally be found placed between greater than and lesser than operators. For example: <variable_input>*

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Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Healthcare does not provide or install wires, conduits, junction boxes, and ducts, as illustrated in this publication.

General Electric Headquarters Architectural Planning must review all CT site plans, preliminary concepts and final working drawings prior to construction or approval.

Contact your local General Electric sales representative or Project Manager of Installation for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

Site ready is a requirement that must be achieved to install a CT product. For your convenience, a site ready visit inspection is performed at least three (3) days prior to the installation date. The site inspection must conclude with a minimum of a conditional pass status to be ready on the requested installation delivery date. Site ready inspections on the delivery date are not acceptable unless prior arrangements have been made.

Pre-Installation and Site Ready Tools:

- Floor template
- Pre-installation checklist
- Site room layouts
- Power and grounding inspection
- Pre-installation support

Section 2.0

Responsibility of Purchaser

2.1 Customer Room Prep Items

The CT air intake is near the bottom of the gantry and draws air in through a filter in the gantry heater assembly. Fine dust, as listed below, clogs the filter and is deposited throughout the gantry, table, operator console and PDU electronics. This fine dust cannot be completely removed and can damage electronic components.

for these reasons, the scanner should be the last item installed in your CT suite area.

- Pre-installation is work necessary to plan and prepare a site for installation of equipment.
- Pre-installation work helps the user (customer) avoid:
 - Application delay and scheduling
 - Surprise siting discoveries
 - Installation confusion
 - Waste of manpower

The following **MUST** be completely finished before installation work can begin for a GE CT scanner:

- Completely finished:
 - Wall painted or have final wall covering
 - Ceiling tiles installed and no remaining ceiling work is required
 - Final floor covering installed with no remaining dust-causing floor work required
 - All room millwork installed as shown on the site print
 - All plumbing work in the CT suite is completed
 - No construction in or around the scan suite AREA that will produce:
 - * Concrete dust
 - * Drywall dust
 - * Ceiling tile dust
 - * Wood sawdust or shavings
 - * Dust tracked into the CT suite area
- Active broadband connection:
 - A completed network connection is required for ALL CT installations.
 - A GE Healthcare network specialist may be required to complete the VCN connection this may take a week or longer to schedule.
- Power available to the A1, with provision for Lockout/Tagout at the A1 disconnect.

If a UPS is ordered, a GE A1 disconnect is required to complete this installation. Refer to



NOTICE

Service Note:

An improperly prepared site (i.e., one that is in a state of construction) can result in increased installation time.

A CT scanner installed in a dirty environment is more prone to contamination, which can result in decreased reliability and increased scanner downtime.

2.2 Purchaser Site Preparation Work

The following list describes many of the items to consider when planning for a system replacement or designing a room for new equipment:

- Determine room dimensions and verify that doorways are large enough for the scanner system.
- Install appropriate conduits and duct work for system cables. If additional components are required in the CT suite, their connection consideration must be determined and completed.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Install an A1 main disconnect.
- Install a power supply of correct voltage output and adequate kVA rating.
- Install local disconnects, including proper over-current protection.
- Install steelwork or other suitable support work for mounting equipment on walls or from the ceiling.
- Camera should be on-site at the time of installation.
- Complete all suite and room alterations and modifications.
- Verify that room shielding is adequate for the system being installed.
- Review structural requirements, including floor vibration, levelness, and thickness.
- Review HVAC requirements, including system regulation and patient comfort.
- Review operational clearances to see if your daily used items fit, such as beds and carts.
- Consider emergency medical equipment.
- Show storage cabinets and sink (if required) on the site print.
- The following contractors, and others, may be required to help confirm that the site meets all installation requirements:
 - Structural Engineer and/or Architect
 - HVAC contractor
 - Electrical contractor
 - Qualified radiological health physicist.
- The project manager of sales will deliver a copy of the pre-installation manual. This work should be completed at least three (3) days prior to delivery.

2.3 Manufacturer's System Level Siting Requirements

The following minimum siting requirements must be met to install a new or replacement system:

- Network communication in place and active
- Meets all scan room regulatory and service requirements
- Meets all minimum scan room structural requirements
- Meets minimum scan room HVAC requirements
- Meets minimum scan room electrical requirements
- Reviewed radiation protection section in the Pre-installation manual
- All in-room items are shown on the final GE Healthcare site print, and the final print is on-site
- No construction in the scan room or neighboring suite areas

This work should be completed at least three (3) days prior to delivery.

2.3.1 Meeting Site Ready Requirements

A site ready visit is suggested for all installations and the visit should take place at least three days prior to the delivery date. The site ready visit is intended to verify that all of the siting requirements are met and the site is ready for installation.

The site ready visit results in a report to the project manager indicating one of the following:

Pass—All required items are present, completed and the site is ready for installation.

Conditional Pass—Eighty percent (80%) of all the tasks are completed and all parties agree that the remaining 20% will be completed by the installation delivery date. If a Conditional Pass is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are complete and that the site is ready for delivery one business day prior to delivery.

Fail—Less than 80% of the tasks are completed and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites should be rescheduled when all unfinished items are completed.

2.3.2 Quick Installs

Quick installations are described as sites with minimum room improvements required. These include, but are not limited to, the following items:

- Existing electrical disconnect device, wire size, and grounds meet all of the above requirements.
- Existing structural items, including floor thickness, meet all of the above requirements.
- Existing HVAC capacity and regulation meet all of the above requirements.
- Existing CT suite meets all of the above regulatory and minimum size requirements.
- Existing facility can accommodate the delivery and meet all of the above delivery requirements.

Quick Installs are subject to the following restrictions:

- Quick Installs must have a new room print that accurately reflects the rooms to be upgraded.
- Existing floor anchors CANNOT be reused unless they are from the same system type and they meet all anchoring requirements.
- New floor anchors must be a minimum of 10 cm (4 in.) from any existing floor penetrations.

Quick Installs typically involve a weekend de-install and room prep completion, with a next business day delivery and install.

2.3.3 Two-Step and Upgrade Installs

A Two-Step Installation is the practice of temporarily installing one CT system in a site with the intention of upgrading the site to a different CT system at a later date.

- For a Two-Step Installation to be considered, the room must meet the minimum room requirements for the upgraded/final project.
- As with any upgrade installation, Two-Step Installations are subject to ALL of the siting requirements imposed by the upgraded/final system. This includes the recommended room size, as well as electrical, structural, and HVAC requirements.
- Two-Step Installations and other upgrades may be done as “Quick Installs.” In this case, all requirements described in [Section 2.3.2](#) (above) also apply.
- It is the customer’s responsibility to check that all requirements are met.
- Sites that do not meet the minimum room requirements for the final product must either upgrade (or enlarge) their room.

2.3.4 Site Ready Inspection Visit (Optional)

The GE Healthcare Project Manager of Installation (GE PMI) will review the site delivery process with you to determine how to best transfer the equipment from the transportation truck to your room. Before the installation, the site must meet all requirements in [Section 2.1](#) and [Section 2.2](#), plus the following items:

- Delivery information
 - Determine delivery route into the scan room
 - Determine if tilt dollies or riggers are required.
 - Determine if elevators, doorways and hallways are adequate for delivery.
 - Determine if floor protection is required.
 - Determine if a tilt bed truck is required for ground delivery and appropriate size truck is ordered.
- Regulatory Requirements
 - Room size meets the minimum requirements
 - Site print is present and accurately reflects the room size and layout
 - No grounded walls are present in the regulatory clearance areas.
 - All regulatory clearances space is met:
 - * Room meets all local codes
 - * Correct cable lengths ordered
- Manufacturer Requirements: As listed in [Section 2.0](#), all requirements are met.
- Purchaser's Site Preparation Work: As listed in [Section 1.0](#), all actions are completed.

Section 3.0 Pre-Installation Checklist

Required Information for Site

Must be completed before the scheduled delivery date

Hospital Name as it appears on the system screens:

Network ID numbers / IP addresses

PACS: _____

Camera: _____

AW: _____

Other - Specify type & ID: _____

Other - Specify type & ID: _____

Camera setup information: _____

AW Direct Connect address: _____

Do you want HIPAA enabled? No___ Yes___

Do you want automatic downloads enabled? No___ Yes___

Table 1-1 Schedule Date Commitments

GE	Cust	Dates
Y N	Y N	
	☐ ☐	Has the project schedule been verified with facilities department, contractor, and GE?
	☐ ☐	Will the committed site-ready date be met?
☐ ☐	☐ ☐	Does the completion date for any/all construction meet or precede the delivery date?
☐ ☐	☐ ☐	Is the Power & Ground survey complete? Date: _____ Hospital contact: _____
	☐ ☐	Site-Ready visit is scheduled. Date: _____
	☐ ☐	Delivery date is scheduled. Date: _____
	☐ ☐	Installation date is scheduled. Date: _____
☐ ☐	☐ ☐	Installation timing: A: Weekdays___ B: Weekend___ C: Quick Install___ If B or C, have all sub-contractors been notified? No___ Yes___
☐ ☐	☐ ☐	Does the delivery and/or installation date need to be adjusted?
	☐ ☐	First-Use date is scheduled. Date: _____
☐ ☐	☐ ☐	Applications/Training dates: On-Site Training Date: _____ Healthcare Institute Training Date: _____

Table 1-2 General Site Planning

GE	CUST	General / Site Requirements
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Have final drawings been approved and distributed to the contractors?
☐ ☐	☐ ☐	Are final drawings “signed off” to approve equipment layout / orientation?
☐ ☐	☐ ☐	Do the actual room dimensions match those on the final drawings?
	☐ ☐	Has the radiologist health physician reviewed and approved the room layout and shielding requirements?
	☐ ☐	Have any additional requirements or questions about the installation been discussed with GE? List: _____ _____ _____ _____
	☐ ☐	Is there a person assigned to review and verify that all installation requirements are met? Name: _____
	☐ ☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 1-3 below.)
	☐ ☐	Has the responsibility of cabling, installing, and interfacing accessories not on the order been discussed?
	☐ ☐	Are all third-party vendors identified, notified and scheduled? (Examples: Netcom, Medrad, etc.)
	☐ ☐	Have all regulatory requirements been met per Component Dimensions , on page 55?
	☐ ☐	Will existing network, broadband, and camera cable drops reach new locations and will they meet the requirements and function with LightSpeed? If not, what are the requirements? List: _____ _____ _____

Table 1-3 References for Specific Site Requirements

Sections for Specific Requirements	
Room Planning - Chapter 4 (page 45)	Floor Loads & Weights - Chapter 6 (page 79)
Radiation Protection - Chapter 4 (page 68)	Delivery - Chapter 7 (page 89)
Environment - Chapter 5 (page 71)	Power - Chapter 8 (page 99)
All work by contractors must be completed before the scheduled delivery date.	

Table 1-4 Equipment Compatibility

GE	Cust	Equipment
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Has the order been reviewed for completeness and compatibility with existing equipment? Typical equipment: Remote monitors ____ AW relocation ____ Cardiac option ____ Injectors ____
☐ ☐		Are interfaces to existing and/or new accessories ordered and planned for accordingly?
☐ ☐	☐ ☐	Have the following peripheral locations been included in the site drawings? EKG monitor ____ Injector control ____ Laser camera ____ UPS ____ 2 nd Monitor ____
☐ ☐	☐ ☐	Will GE Healthcare provide additional services per contract negotiations?
☐ ☐		Are correct length cables on order?

Table 1-5 Network Connections

GE	Cust	Network Installation and Setup
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Have IP addresses and Host Names been obtained? No ____ Yes ____
☐ ☐	☐ ☐	Will a network camera be used? No ____ Yes ____
☐ ☐	☐ ☐	Mandatory: Is the network installed, are network jacks installed, and is the entire network tested?
☐ ☐	☐ ☐	Mandatory: Broadband VPN installed/setup?
☐ ☐	☐ ☐	Mandatory: Are network software options ordered ____ HIS RIS option ____ DICOM print ____ AW ____
	☐ ☐	Optional: Has modem option ordered? ____ (Requires a site escalation)
☐ ☐	☐ ☐	Optional: Is the LightSpeed service telephone line identified and installed for InSite? (<i>Electrical, mechanical, etc.</i>)

Table 1-6 Miscellaneous Tasks

GE	Cust	Other
Y N	Y N	<i>Must be completed before the scheduled delivery date</i>
	☐ ☐	Arrangements made in the schedule to allow adequate time for remodeling, if required (such as wall, floor, or ceiling repair work, painting, other cosmetic finishes).
	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to the installation of the new equipment?
☐ ☐	☐ ☐	Is de-installation of existing equipment required? No__ Yes__ Removal date _____
☐ ☐	☐ ☐	Is there a trade-in of existing equipment? No__ Yes__ GoldSeal _____
☐ ☐	☐ ☐	Delivery route identified and verified with the proper hospital personnel? No__ Yes__ Elevators and doors checked for size and weight constraints? No__ Yes__
☐ ☐	☐ ☐	Have appropriate arrangements been made with traffic for delivery? No__ Yes__
☐ ☐	☐ ☐	Will acceptance/performance testing or bio-medical testing be required? No__ Yes__ Date: _____
☐ ☐	☐ ☐	Are trash and/or recycling bins available for the removal of papers, boxes, etc. during the installation? No__ Yes__

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Chapter 2

Pre-Installation Overview

Before installing a LightSpeed system, all pre-installation requirements must be complete.

- | | |
|--|--|
| • Chapter 1, Section 2.3.4 | Site Ready Inspection Visit |
| • Chapter 4, Section 5.0 and Section 6.0 | Structural Requirements; Network Connections |
| • Chapter 4, Section 6.1 | US Broadband Process Overview |
| • Chapter 4, Section 7.0 | Radiation Protection |
| • Chapter 5, Section 1.0 , Section 3.0 | HVAC Requirements |
| • Chapter 8, Section 5.0 | Power Audit |
| • Chapter 9, Section 3.0 | Interconnect Runs, Wiring, and Cable |

Site-specific items must be verified before the installation can begin.

Section 1.0

Dust/Dirt Contamination

The LightSpeed system (consisting of: console, PDU, table and gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.

Note: **Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.**

Section 2.0

Chemical Contamination

Never install wet film processors in the same room as the scanner, due to the possibility of chemical contamination of the LightSpeed series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consider the location of this equipment and chemical fumes relative to human contact, as it relates to locating this equipment and chemicals in the control room.

Section 3.0

Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin. LightSpeed systems can only be installed on a 10 cm (4 in.) concrete floor.

Section 4.0 Broad-band

For information on broadband requirements, refer to [Chapter 4, Section 6.1, on page 67](#).

Section 5.0 Phone Line (for optional modem)

Two phone lines must be installed at or near the operator console and be operational prior to installation.

- One analog line (for modem use)
- One voice line

Section 6.0 Review

The LightSpeed systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. The table has five (5) leveling pads.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the supplied template (P/N 2265473) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the two (2) floor templates.
- 2.) Start with the gantry template—align it per the GE print.
- 3.) Place the table template over the top of the gantry template.
- 4.) Align the scan and table centerlines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 5.) Check the level of the floor (See [Figure 2-1](#)) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

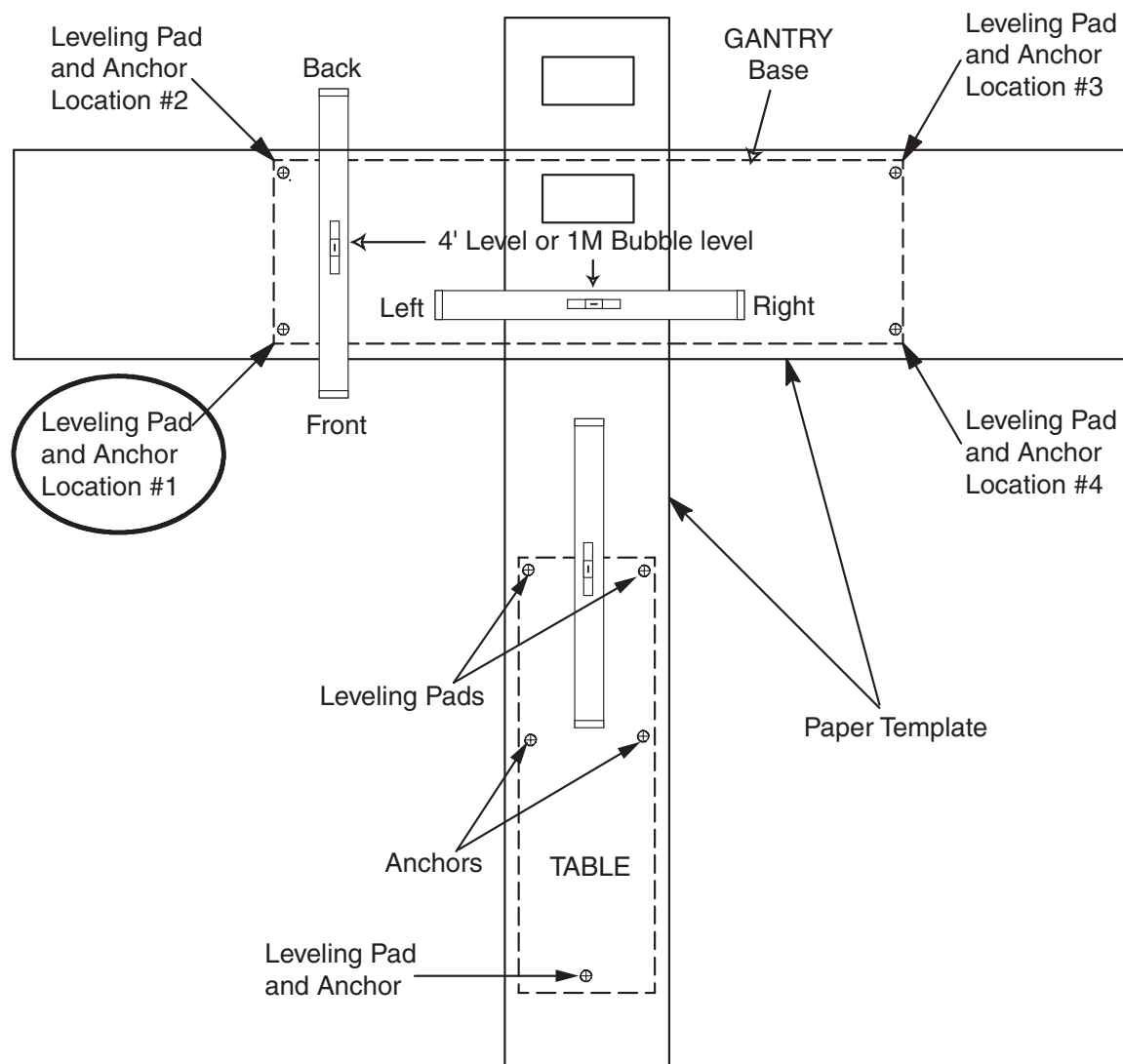


Figure 2-1 Hole Locations

FLOOR LEVELNESS SPECIFICATION

3 mm (1/8 in.) over 3048 mm (10 ft.)

This should be measured on the template over the table/gantry area, as shown in [Figure 2-1](#), above.

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Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 3.X systems. Contact your local GE Healthcare Sales force for a complete list of all system options or visit us at <http://www.gehealthcare.com>. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION
Uninterruptible Power Supplies. B7999PS	"Powerware" Un-interruptible Power Supply
International Dolly Set B7850LD	For International customers, if dollies are required.
Rear Cable Cover B7850RC	Optional rear cable cover for LightSpeed 3.X upgrades
SmartScore Option B7864CB	EKG Monitor and Recording Device
Performance Network Kits K9000L	6 Node, 10/100 Mbit Auto Sensing
Bar Code Reader B7540RB	LightSpeed 4.X Bar Code Reader
Seismic Kit R4390JC	Earthquake zones
Modem B7700MG	
Long Cable Set B7850LL	Optional Long Cable Set

Table 3-1 System Options Catalog (Part) Numbers

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0

Regulatory and Service Clearances

1.1 Regulatory Clearances

MINIMUM CLEARANCES UNDER U.S. FEDERAL REGULATIONS AND NATIONAL STANDARDS:

29 CFR 1910 (OSHA),

NFPA 70E (STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE)

NFPA 101 (LIFE SAFETY CODE):

Figure 4-1 is a map of clearance requirements for U.S. regulatory compliance. See clearance tables on the following pages for detailed dimensional clearances. Please note: all systems installed in the United States must comply with all Federal and local regulations. For installations outside the United States, country-specific or other local regulatory clearance requirements must be met.

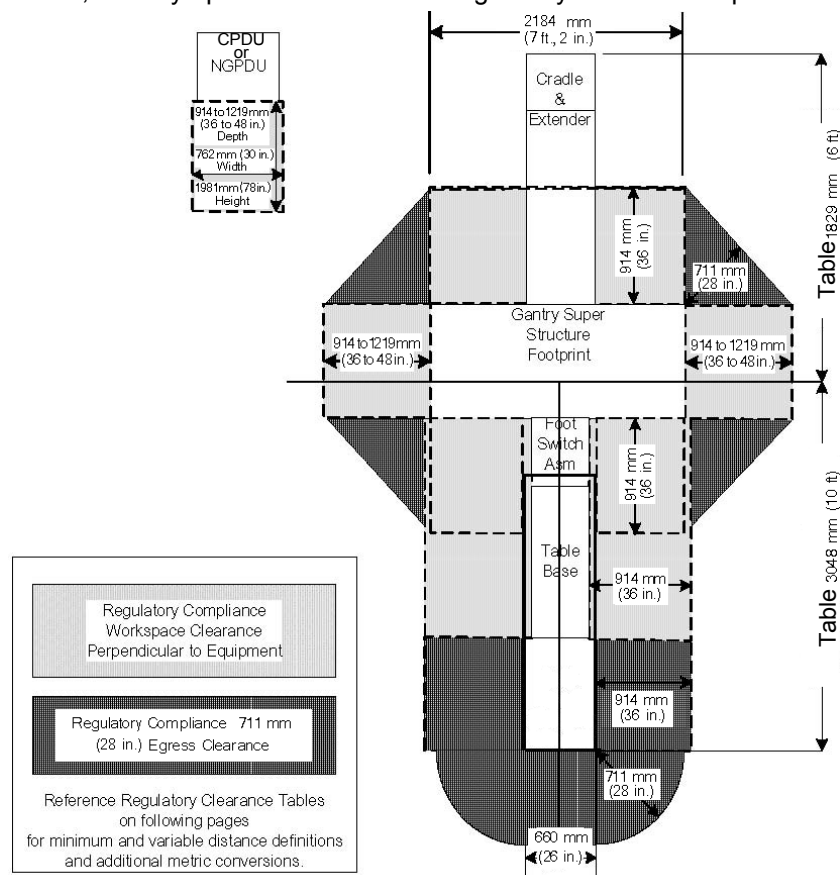


Figure 4-1 Regulatory Clearance Requirements for LightSpeed System Configurations

Note: See [Section 2.0](#) for Service Clearances

1.2 Regulated Minimum Working Clearance by Major Subsystem

- Consult your local GE Project Manager of Installation for your appropriate room specifications.
- Requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.
- Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.
- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.
- Where possible, the cables should enter the gantry from the rear, utilizing the rear cable cover assembly. Alternate cable entry is possible at the center of the gantry (refer to the installation template for your scanner).

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access	N/A No exposed live part hazards	

Table 4-1 Console

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (Front of PDU)	914 mm (36 in.)	1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1067 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 30 inches (762 mm) min or the width of the equipment, whichever is greater
Head Clearance	1981 mm (78 in.)	The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s). 78 inches (1981 mm) or height of equipment, whichever is greater

Table 4-2 CPDU and NGPDU

- For the Gantry and Table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1067 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.

Table 4-3 Gantry

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min. or the width of the equipment, whichever is greater.

Table 4-3 Gantry

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (Table Head)	N/A	
Direction of Service Access (Table Sides)	914 mm (36 in.)	*Can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). Signed document must be on file within GE.
Direction of Service Access (Table Foot)	711 mm (28 in.)	457 mm (18 in.) minimum for Front Gantry Cover removal only if unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min. or the width of the equipment, whichever is greater.

Table 4-4 Table

1.3 Terms and Definitions

EGRESS

Egress is the path of exit from within any room. U.S. regulatory requires a minimum of 711 mm (28 in.) of continuous and unobstructed space including trip hazards along the path of exit.

WORK SPACE

Work Space is the dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. U.S regulation is minimum of 914 mm (36 in.). Additional conditions can increase the minimum requirement. FCT defines this as the envelope of the component superstructure. For the NGPDU it is with the front panel removed. For the gantry and table, it is with the patient or external covers removed.

SERVICE ACCESS WIDTH

Service Access Width is the width of the working space in front of the equipment, a minimum of 762 mm (30 in.), or the width of the equipment, whichever is greater.

HEAD CLEARANCE

Head Clearance is the height dimension of "Work Space." The height of the workspace is measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s), 1981 mm (78 in.) or height of equipment, whichever is greater.

GROUNDING WALL

A Grounded Wall is any wall that can be electrically conductive to earth ground. Masonry, concrete, or tile are considered conductive. Additional commonly found aspects of a wall should also be considered as grounded. The following is not an all-inclusive list:

- Medical Gas ports
- Metal door and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 disconnect panel
- Equipment Emergency Off panels
- Industrial equipment, such as air conditioners and vents
- Expansion joints

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks

MINIMUM

Minimum refers to the lowest limit permitted by law or other authority.

DIMENSIONS AND CLEARANCES

Dimensions and Clearances consist of, or represent, the lowest possible amount of degree for freedom permissible for equipment siting. This relationship must meet all safety, service, and regulatory requirements to be acceptable.

1.4 Additional Regulatory Clearance Information

1.4.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout, as shown on your site print, shall meet all regulatory requirements as described in the installation manual. Additional room components, such as cabinets, reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

1.4.2 Operational Caution

In a minimum room layout 5867.4 mm x 4114.8 mm (19 ft. 3 in. x 13 ft. 6 in.), the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall.

1.4.3 Egress Clearance

Egress requires a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 457 mm (18 in.) of clearance between the back of the table, with a continuous width of 3200 mm (126 in.), (1600 mm (63 in.) on each side of the table center line), to any obstruction, so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

1.4.4 System Specifications

System Configuration	Preferred Room Size	Typical Room Size	Minimum Room Size
LS 16	6705.6 x 4267.2 mm ¹ (22 ft. x 14 ft.)	6096 mm x 4267.2 mm ¹ (20 ft.x 14 ft.)	5867.4 mm x 4114.8 mm (19 ft. 3 in. x 13 ft. 6 in.)
¹ Same Regulatory requirements apply			

Table 4-5 System Specifications (Pro16)

1.4.4.1 Preferred Room Size

The “preferred room size” configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add mill work, while meeting all regulatory requirements. This room is compatible with most two-step future installations, and a 7.X VCT.

1.4.4.2 Typical Room Size

The “typical room size” configuration allows for some future upgrades. It has sufficient workspace, but limited space to add mill work and meet all regulatory requirements. This room is compatible with two-step future installations, and a 5.X Pro16 or 5.X RT.

1.4.4.3 Minimum Room Size

The “minimum room size” configuration allows for no future upgrades. It has limited workspace and limited in-room mill work, but meets all regulatory requirements. This room is compatible with two-step future installations, and a 5.X Pro16 or BrightSpeed.

1.5 How to Measure

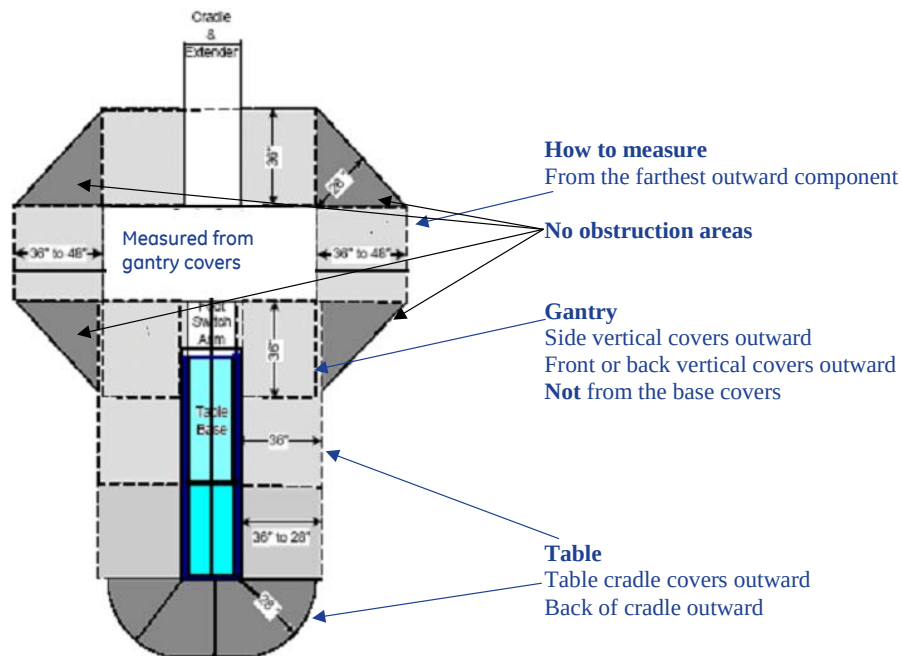


Figure 4-2 How to Measure a Room for Clearance

1.6 Preferred and Minimum Room Size and Requirement Layouts

Consult your local GE Sales and Service Representative about your specific needs.

Note: **Limited Access is not available for this product.**

Figure 4-3 and Figure 4-4 show room layouts, with and without in-room cabinets. You need to be aware of locations for med gas, surface duct-work, or other items that make a grounded wall.

Note: Your room layout may meet the preferred or typical room requirements but look different than the examples. Equipment shown in these details may not represent the product you are siting. Contact your sales person to have a detailed room layout completed for your site.

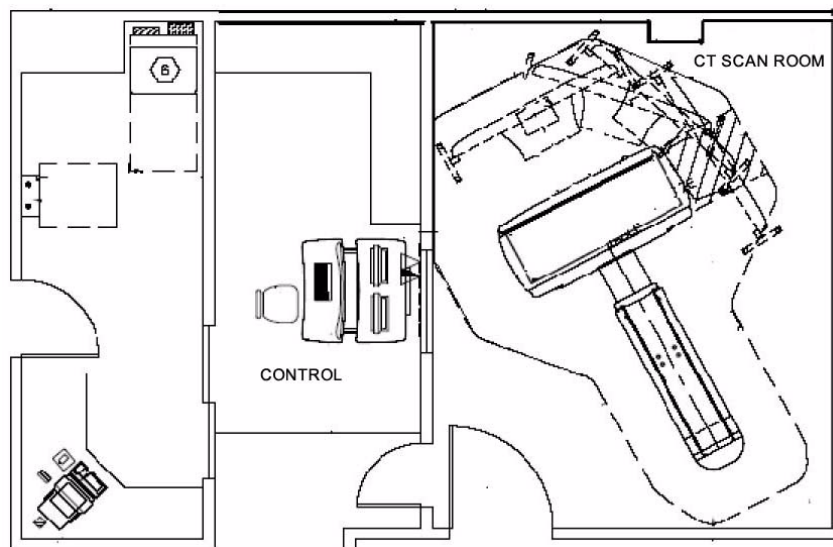


Figure 4-3 Room layout without cabinets

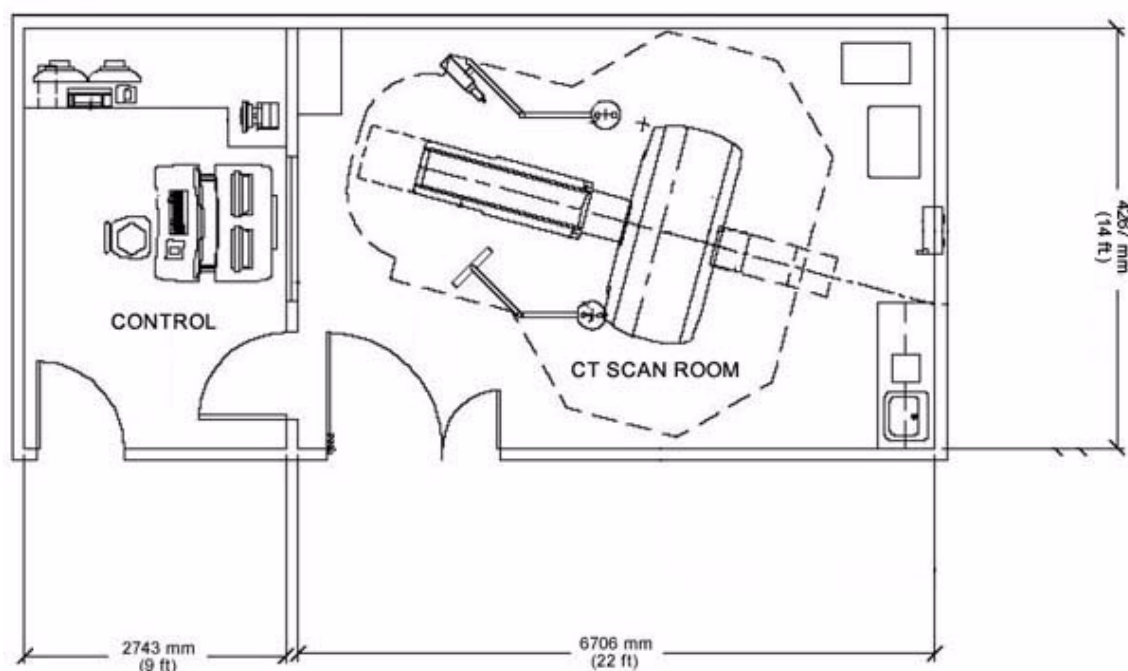


Figure 4-4 Room layout with cabinets

Section 2.0 Service Clearances

Servicing the CT System can be safely performed within the regulatory envelopes defined in [Section 1.1](#) of this chapter. However, sufficient space must be maintained to remove the covers from the system. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. One Service Engineer can accomplish this.

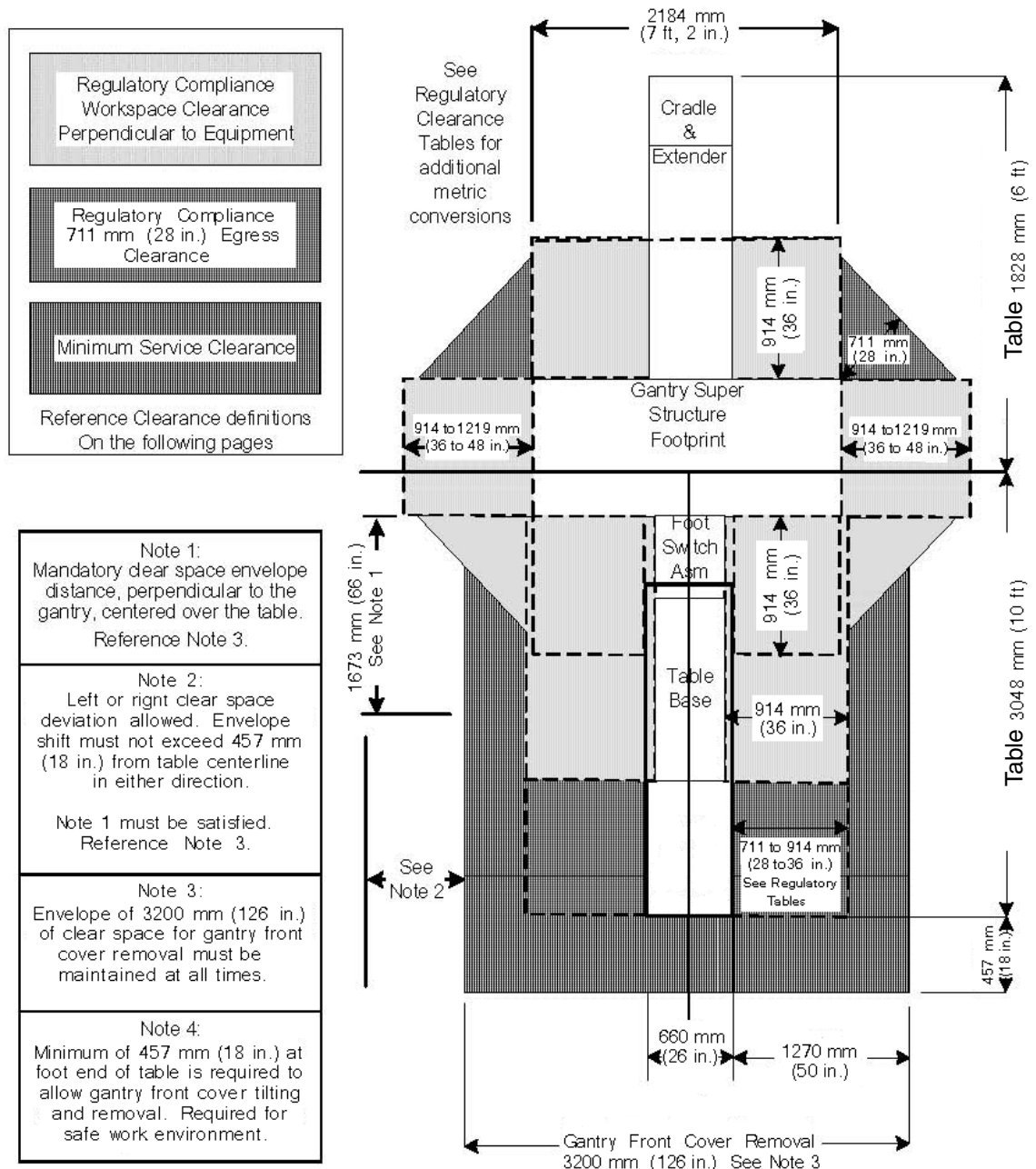


Figure 4-5 Minimum Service Clearance

2.1 Service Clearances for Single Service Engineer

- Gantry front cover removal requires the use of the Tilting Cover Dollies. These dollies allow the Service Engineer to separate the cover from the gantry, tilt the cover 90 degrees, roll the cover to the foot end of the table, and then tilt the cover an additional 90 degrees, such that the front cover is now upside down relative to the normal system-mounted condition. [Figure 4-5](#) illustrates the minimum clear space required to achieve this operation of 3200 mm (126 in.). The gantry front cover must be removed to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover with service dollies installed, requires a width of 2388 mm (94 in.) and a depth of 584 mm (23 in.) of clearance for removal as shown in [Figure 4-5](#). Sufficient space must be calculated to move the cover either straight back or to one side of the table to satisfy the minimum service clearance shown in [Figure 4-5](#). This means the rear cover cannot violate the workspace on the rear, or on either side of the gantry.
- If gantry service requires both the front and rear covers be removed, then these covers must be positioned within the room in such a manner as to not violate the egress clearances on any side of the gantry. This may necessitate removing the covers from within the suite. This should be discussed with the customer and provisions made to accommodate this potential event.
- A single Service Engineer can safely perform servicing of the table. Sufficient clear space must be available to maintain egress clearances when the table covers or cradle are removed.
- In your room layout design, service personnel must have a clear and unobstructed access to the gantry tube change area for all major component replacements. Be aware of cabinet placement and how surface floor ducts are used in the room configurations. These components must be able to reach the service area without lifting or rigging by one service engineer. Major components include:
 - CT X-ray tube in crate
 - high voltage tank in crate
 - slipring in crate
 - detector assembly

2.2 Power Distribution Unit (CPDU and NGPDU) Service Clearance

Positioning of this component must be considered for regulatory compliance as defined in [Section 1.1](#), Regulated Minimum Working Clearance by Major Subsystem.

2.3 Console Service Clearance

The operator console is not a hazard for exposed live parts. However, a minimum working space depth of 1219 mm (48 in.) and full width of the operator console must be maintained at all times for service activity. Additionally, sufficient space must be provided for repositioning the operator console, and side clearance for rear service access. See [Figure 4-5](#) for a typical control room layout.

2.4 Operator's Console/Control Room Layout

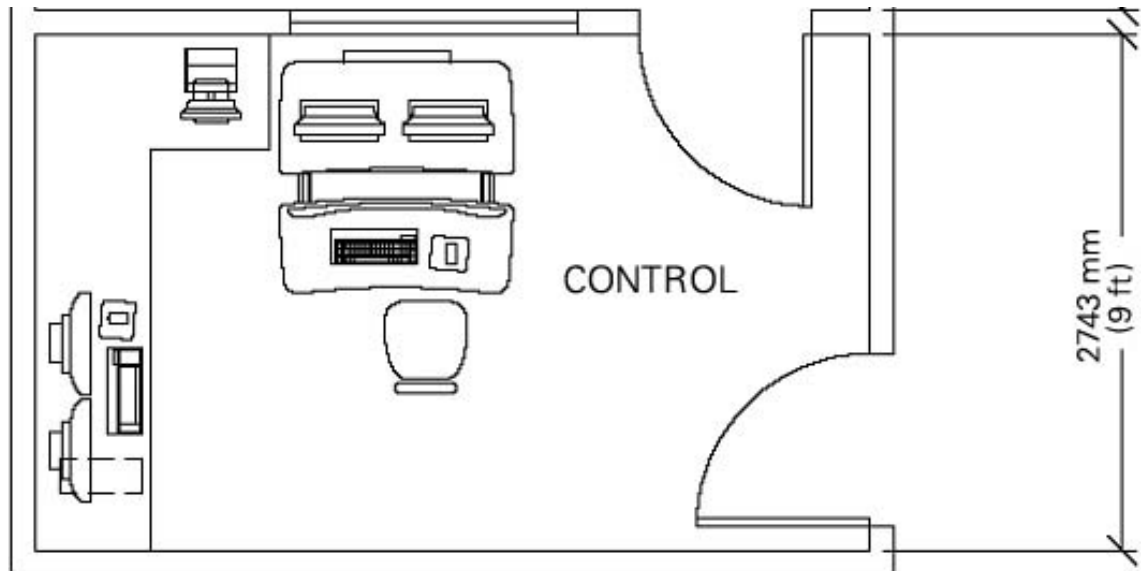


Figure 4-6 Typical Control Room Layout

- The control room must provide a suitable operating environment for the operator console electronic, and operator working comfort.
- Do not dismantle, remove, or rearrange operator console components in configurations other than as shipped.
- If operationally possible, the monitor desktop and user desktop components may be removed and placed on a counter top, providing the cable lengths shipped are not altered or changed. The operator console cabinet then can be mounted remotely, provided the cooling requirements are met. Maintain 152 mm (6 in.) on all sides, and venting is required.
- Provide a suitable work area, which is within reach of the operator console, for placement of the injector control. Injector controls differ in dimensions, depending on the brand selected.
- A PACS, workstation, image printer, or filming device are often placed in the operator console control room area, and sometimes may be directly linked to the operator console.
- Additional components, although linked via network or ethernet cable, are not powered from the CT operator console.
- Consider additional room power and network connection when reviewing the operator console work space.

Section 3.0

Component Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	MM	IN.	MM	IN.	MM	IN.
Systems LS16						
Gantry	2200	86.6	1000	39.4	1895	74.6
Table	617	24.3	2387	94		
Power Distribution Unit PDU NGPDU	762	30	559	22	1270	50
	700	27.6	550	21.7	1062	41.8
Operator's console/computer	1219	48	991	39	85	33.5
Powerware UPS	762	30	406	16	711	28
Workstation	419	16.5	409	16.1	79	3.1
Monitor	495	19.5	483	19	471	18.6
A1	508	20	203	8	787	31

Table 4-6 Dimensions of Components

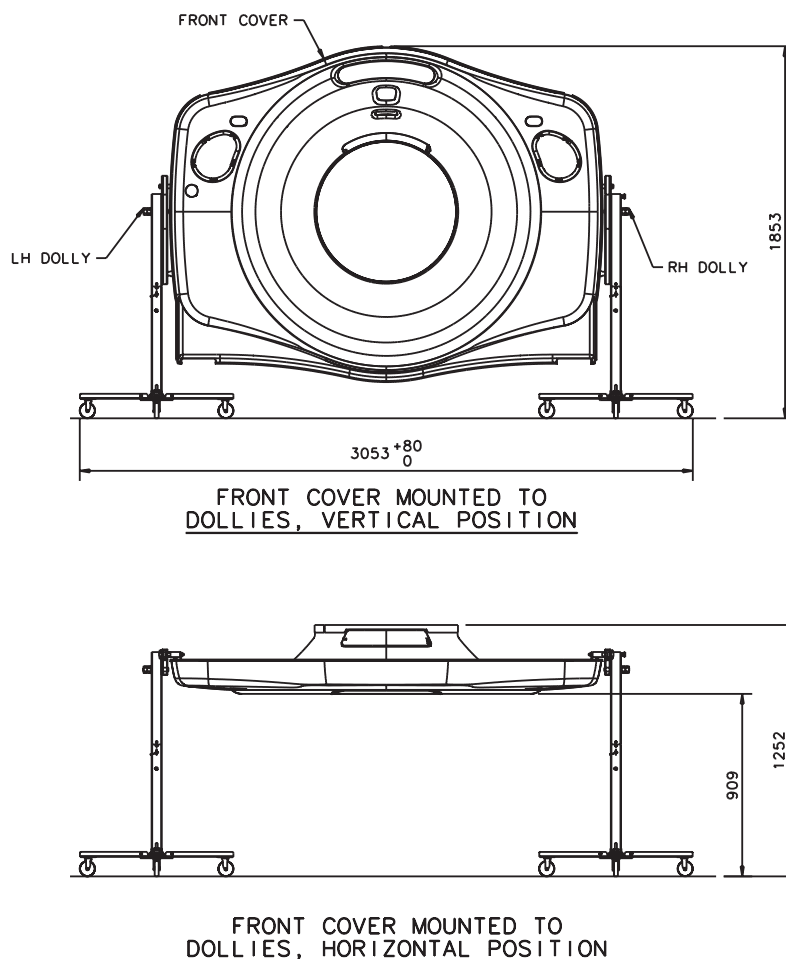


Figure 4-7 Gantry Front Cover with Service Dolly Dimensions

3.1 Storage Cabinet

A storage cabinet should be used to store all supplied service equipment (see [Table 4-7](#) for an equipment list). The recommended size for this storage cabinet is 54.5 kg (120 lb.), and 457.2 mm Deep x 914.4 mm Wide x 1066.8 mm High (18 in. D x 36 in. W x 42 in. H), which should be located in the scan room suite area for easy service access. Refer to [Figure 4-3](#) or [Figure 4-4](#).

ITEM	SIZE	WEIGHT (TOTAL)
QA Phantom (water filled)	20 x 15 cm (7.9 in. X 5.9 in.)	5.4 kg (12 lb)
20CM Phantom	20 x 7 cm (7.9 in. X 2.8 in.)	5.9 kg (13 lb)
48CM Phantom	48 x 7 cm (18.9 in. X 2.8 in.)	11.3 kg (25 lb)
Phantom Holder	25 x 25 cm (9.8 in. X 9.8 in.)	3.6 kg (8 lb)
Gantry Display	62 x 13 cm (24.4 in. X 5.1 in.)	1 kg (2 lb)
FE Box (Purple)	30 x 38 x 30 cm (11.8 in. X 15 in. X 11.8 in.)	6.8 kg (15 lb)
Rear Cover Dollies	158 x 82 cm (62.2 in. X 32.2 in.)	11.3 kg (25 lb)
Front Cover Dollies	85 x 20 cm and 85 x 15 cm (33.5 in. X 7.9 in. and 33.5 X 5.9 in.)	15.9 kg (35 lb)
Install Support Kit (box)	30 x 30 x 38 cm (11.8 in. X 15 in. X 11.8 in.)	9.1 kg (20 lb)
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm (30.3 in. X 3.2 in. and 15 in. X 5.9 in.)	9.1 kg (20 lb)
Gantry Control Pad	27 x 20 cm (10.6 in. X 7.9 in.)	1.8 kg (4 lb)

Table 4-7 Equipment to be stored in storage cabinet

3.2 Advantage Windows Workstation (AWW)

Refer to *Advantage Windows Pre-Installation Manual* (2111833) and *Advantage Windows Service Manual*, including installation (2111831-100).

3.3 Replacement Component Dimensions

The following are major system components that will be expected to be field replaceable on all *fixed* or *non-fixed* systems.

3.3.1 X-Ray Tube

The X-ray tube is shipped in a wooden box. Installers must have an unobstructed path to move this box to the front side of the gantry, and also have working room on the right side of the gantry. Installers must have room to remove the front and rear covers using the current cover dollies. Installers must have a ceiling height of 2.13 m (84 in.) to operate a hoist and boom, installed on the right front side of the gantry.

The X-ray tube measures 736.6 mm H x 711.2 mm W x 1219.2 mm L (29 in. H x 28 in. W x 48 in. L, and weighs 125 kg (275 lbs)

Note: The box with handles extended is 1829 mm (6 ft.) long.

3.3.2 High Voltage Tank

The High Voltage Tank is shipped in a wooden box. Installers must have an unobstructed path to move this box to the front right side of the gantry, and also have working room on the right side of the gantry. Installers must have room to remove the front and rear covers using the current cover dollies. Installers must have a ceiling height of 2.13 m (84 in.) to operate a hoist and boom, installed on the right front side of the gantry.

Single	584.2 mm H x 406.4 mm W x 355.6 mm L (23 in. H x 16 in. W x 14 in. L)	15.9 kg (35 lbs)
Dual	584.2 mm H x 685.8 mm W x 431.8 mm L (23 in. H x 27 in. W x 17 in. L)	31.8 kg (70 lbs)

3.3.3 Detector

The detector is shipped in a wooden box. It must have an unobstructed path to move to the front right side of the gantry, and also have working room on the gantry right side. There must be room to remove the front and rear covers using the current cover dollies. There must be a ceiling height of 2133.6 mm (84 in.) to operate a hoist and boom installed on the gantry right front side.

The Detector box measures: 431.8 mm H x 914 mm W x 1397 mm L (17 in. H x 36 in. W x 55 in. L) and weighs 40.8 kg (90 lbs.).

3.3.4 DAS

The DAS is shipped in a box. It must have an unobstructed path to move to the front of the gantry, and also have working room around the gantry. There must be room to remove the front and rear covers using the current cover dollies. There must be a ceiling height of 2133.6 mm (84 in.) to operate the hoist and boom installed on the gantry right front side.

The DAS box measures: 685.8 mm H x 406.4 mm W x 685.8 mm L (27 in. H x 16 in. W x 27 in. L), and weighs 18.1 kg (40 lbs.).

3.3.5 Slip Ring

The Slip Ring is shipped in a wooden box. It must have an unobstructed path to move to the front right side of the gantry, and also have working room on the gantry right side. There must be room to remove the front and rear covers using the current cover dollies. There must be a ceiling height of 2133.6 mm (84 in.) to operate a hoist and boom installed on the gantry right front side.

The Slip Ring box measures: 1524 mm H x 1524 mm W x 254 mm D (60 in. H x 60 in. W x 10 in. D) and weighs 58.1 kg (128 lbs.).

3.3.6 Gantry Motor Assembly

The Gantry Motor Assembly is shipped in a box. It must have an unobstructed path to move to the front of the gantry, and also have working room around the gantry. There must be room to remove the front and rear covers using the current cover dollies. There must be a ceiling height of 2133.6 mm (84 in.) to operate a hoist and boom installed on the gantry right front side.

The Gantry Motor Assembly box measures: 431.8 mm H x 304.8 mm W x 762 mm D (17 in. H x 12 in. W x 30 in. D), and weighs 20.4 kg (45 lbs.).

3.4 Table and Gantry

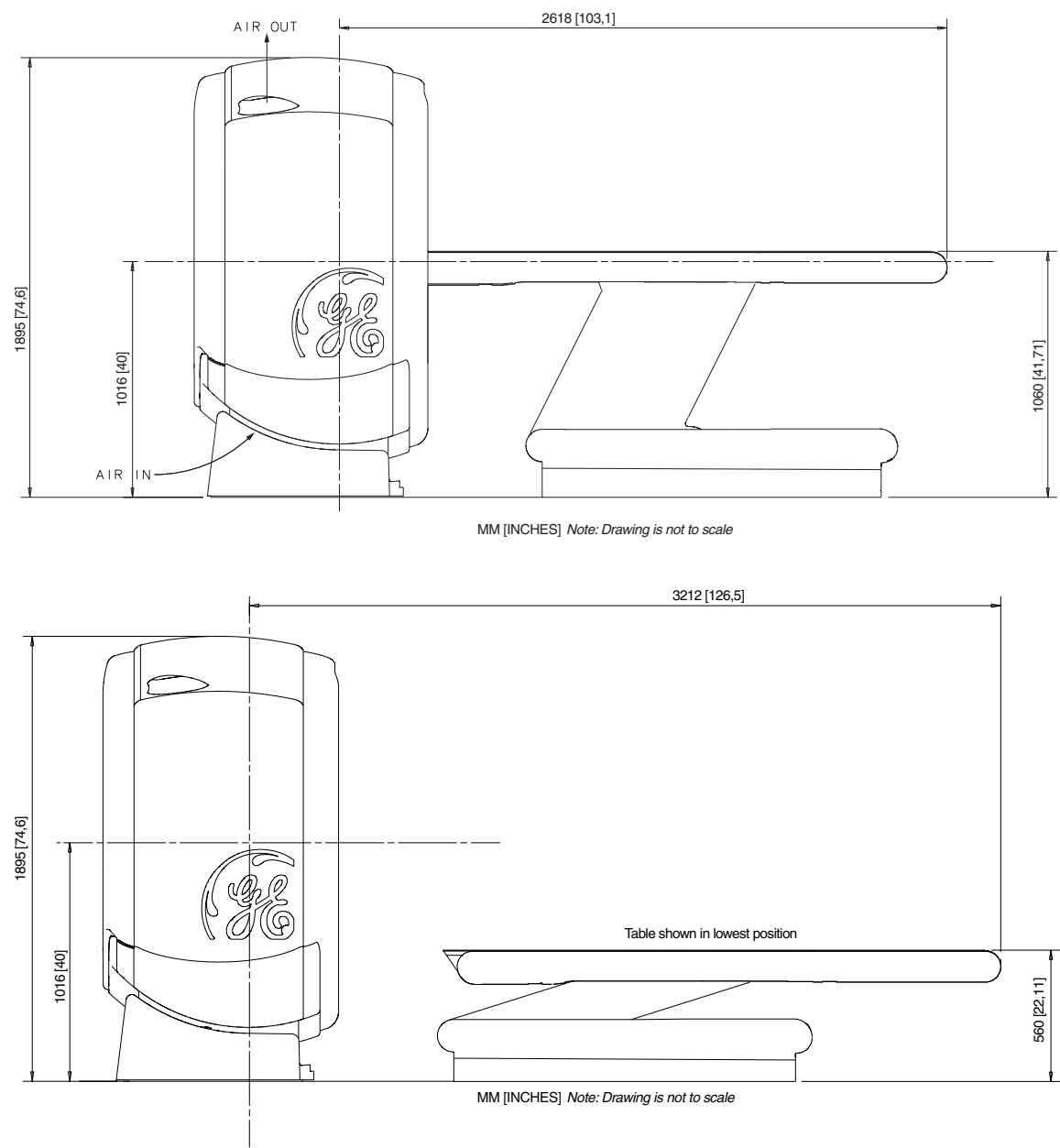


Figure 4-8 Table and Gantry (Side View)

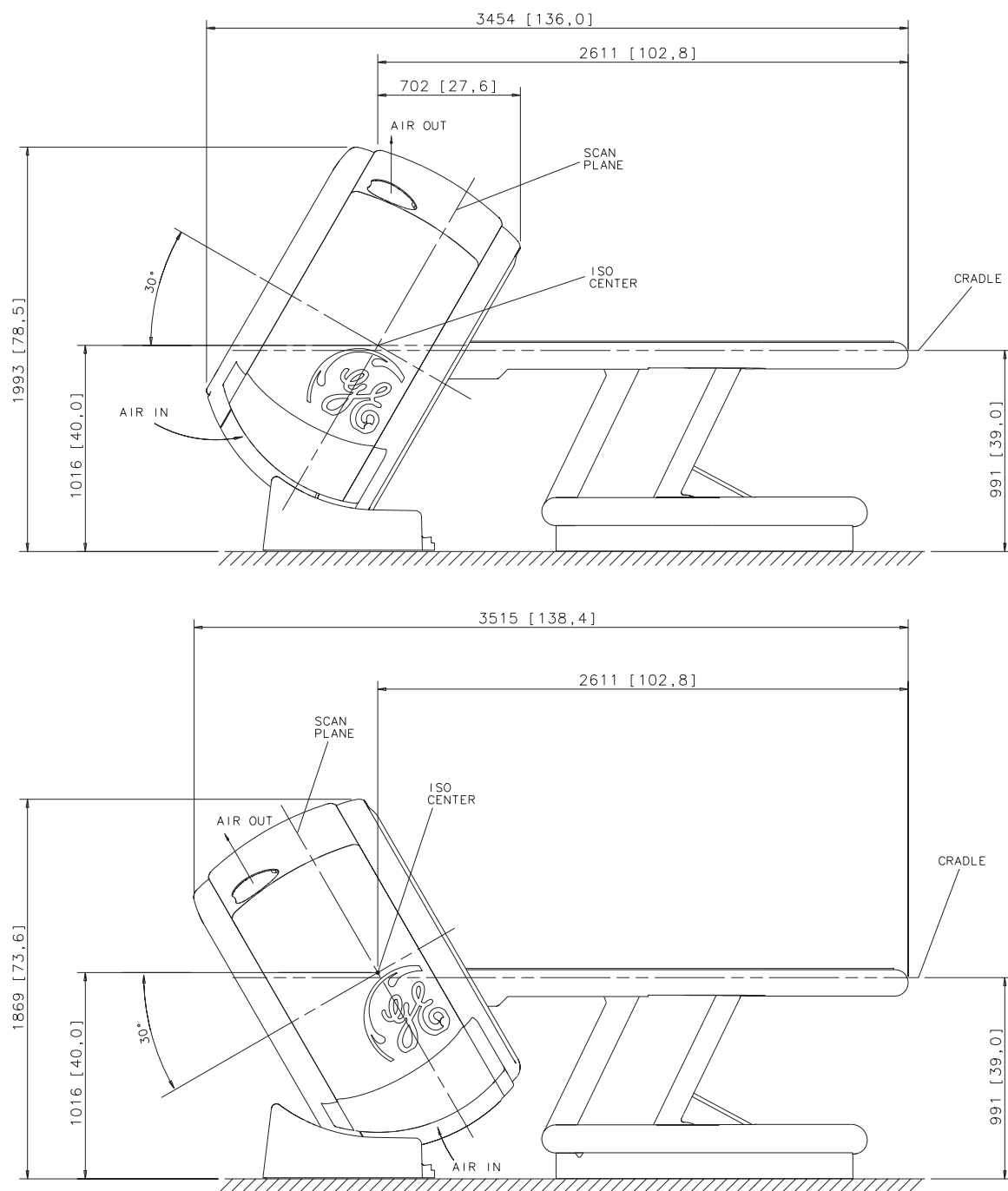


Figure 4-9 Gantry shown tilted +30° (top) and -30° (bottom)

3.5 Power Distribution Unit

3.5.1 Compact Power Distribution Unit (CPDU)

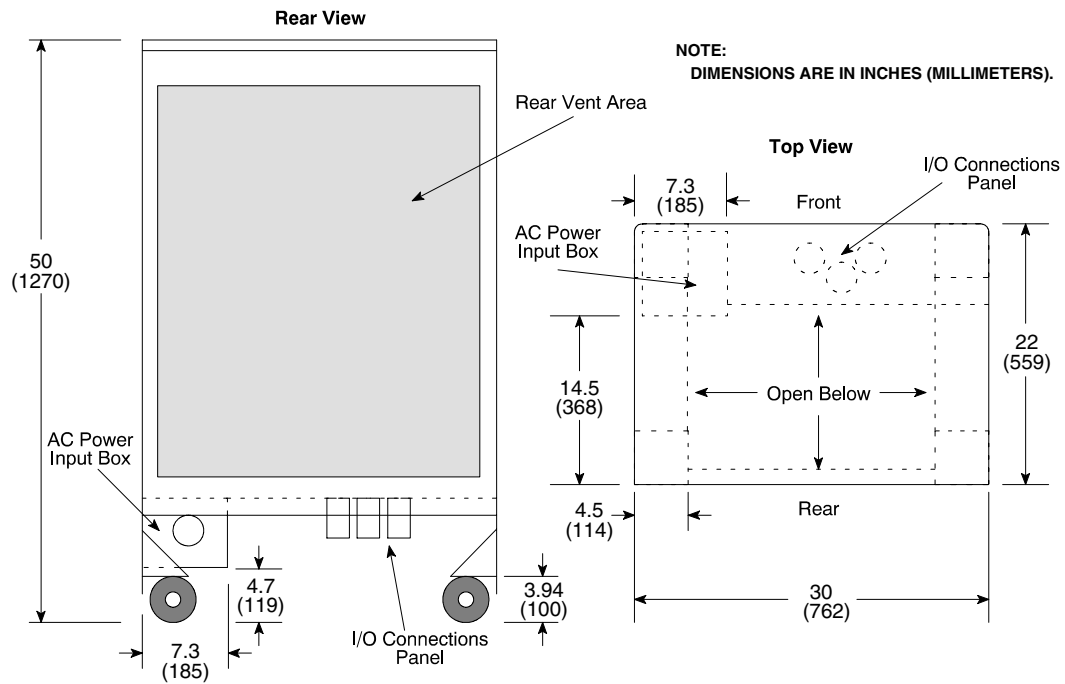


Figure 4-10 Compact Power Distribution Unit (CPDU)

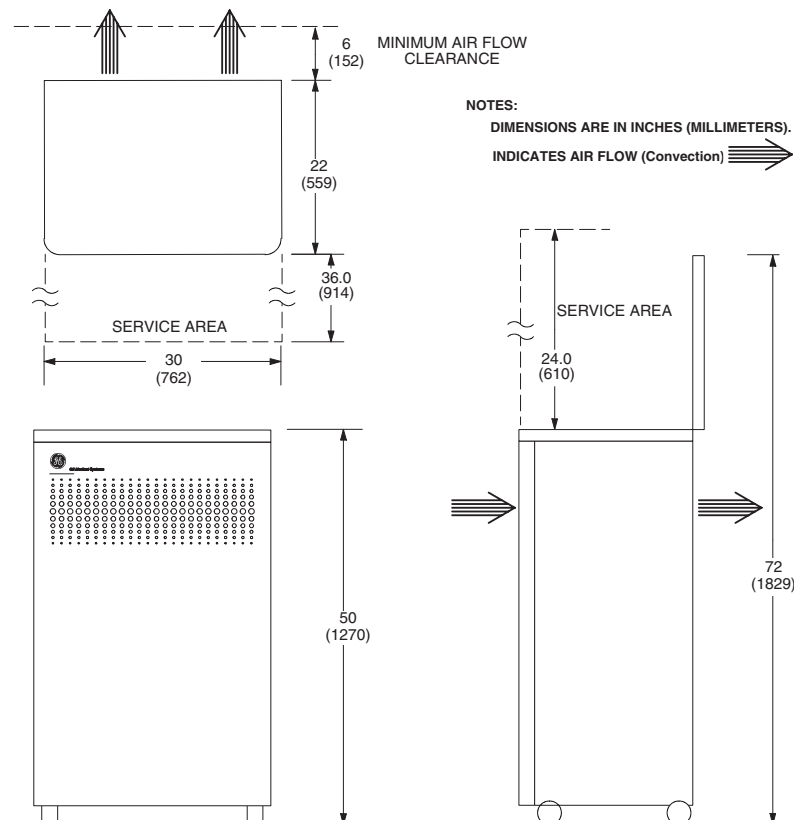


Figure 4-11 Compact Power Distribution Unit (CPDU)

3.5.2 Power Distribution Unit (NGPDU)

NOTE:
DIMENSIONS ARE IN INCHES (MILLIMETERS).

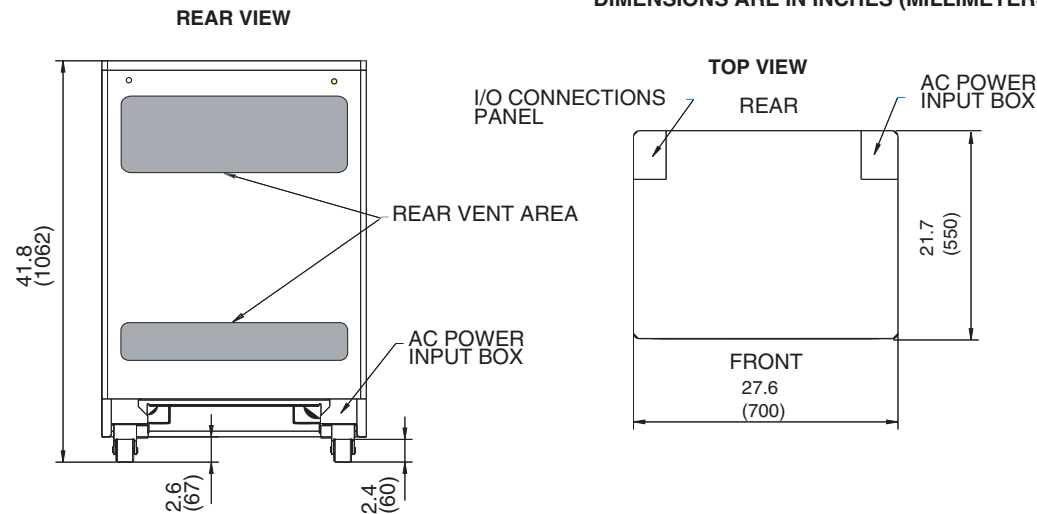
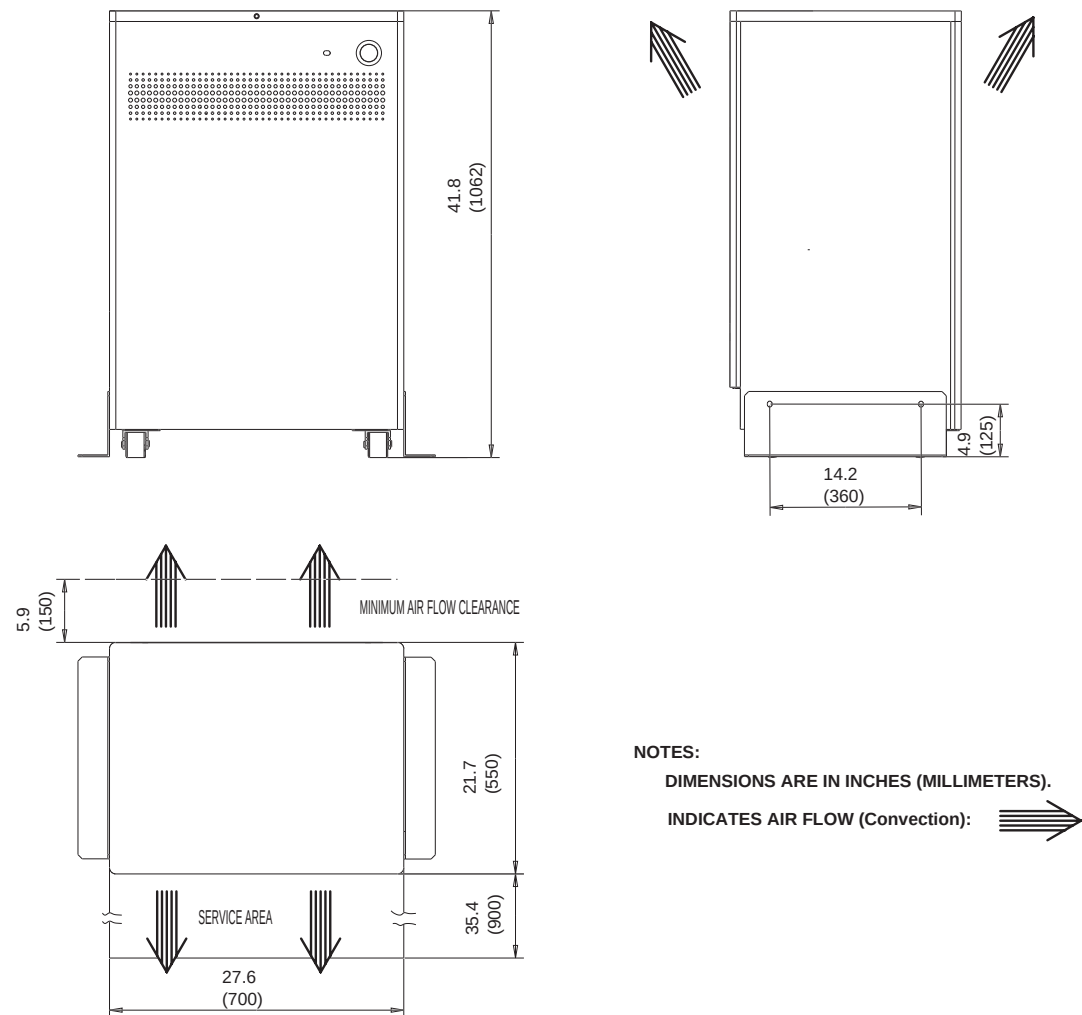


Figure 4-12 Power Distribution Unit (NGPDU)



NOTES:
DIMENSIONS ARE IN INCHES (MILLIMETERS).
INDICATES AIR FLOW (Convection):

Figure 4-13 Power Distribution Unit (NGPDU)

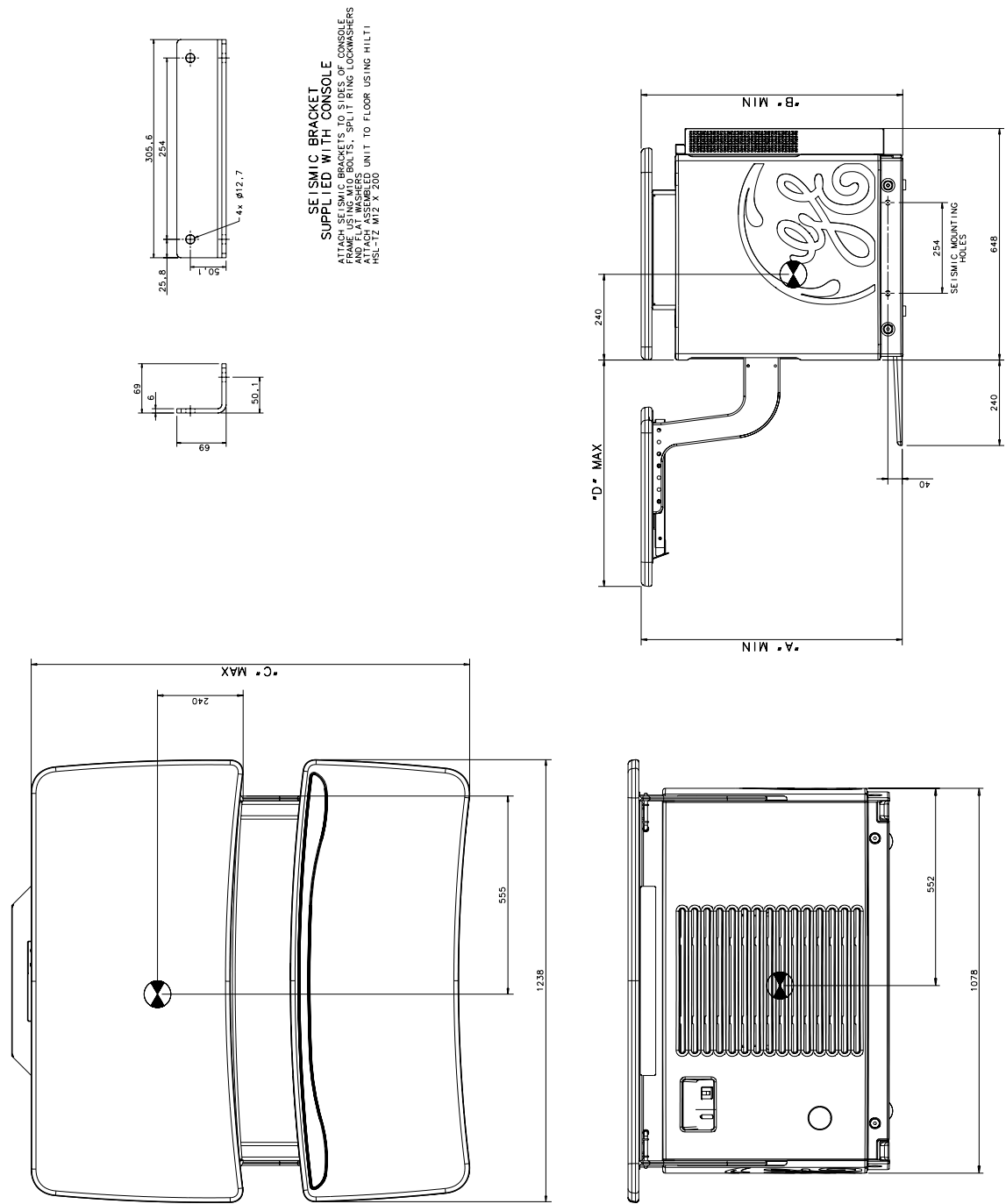


Figure 4-14 Operator's Console

Section 4.0

Minimum Dimensions and Clearances

SYSTEM OPERATION	MM	INCHES
Ceiling pedestal mount (optional) (lowest point to floor injector or monitor)	2439 mm	96 in.
Finished ceiling to floor (recommended)	2743 mm	108 in.
Finished ceiling to floor (minimum)	2439 mm	96 in.
Back of console to wall	152 mm	6 in.
Back of PDU to wall	152 mm	6 in.

Table 4-8 Minimum Dimensions and Clearances

4.1 Injector Control

The customer needs to provide a suitable work area for placement of the injector control, which is within reach of the operator console.

Route the wall-mounted, ceiling-mounted, and pedestal unit cables from the gantry area to the operator console area. The supplied cable is 15.24 m (50 ft.) in length. Injectors require AC power that is not supplied by the CT system.

Note: Do not route injector cables with the system cables.

Mounts are available in different configurations and lengths. Refer to injector documentation for detailed installation instructions.

Section 5.0 Structural Requirements

5.1 Table and Gantry Mounting Requirements

 **WARNING** **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

 **NOTICE** It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements.
GE is not responsible for methods other than those listed.

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), and [Figure 6-5](#). Refer to [Chapter 6](#) for additional details of floor loadings, component weights, and gantry and table installation and anchoring.

Anchor the gantry and table to the floor by a means that maintains their relative alignment and meets applicable building and other local codes, including seismic structural mounting requirements.

The floor structure must be capable of withstanding the occupied weight of the table and gantry, and the individual contact area loading of these components.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting, which will slowly yield over a period of time and disturb the alignment of the table to the gantry.

Consider factors that could cause misalignment between the gantry and table due to floor sag. The cradle can potentially carry a 205 kg (450 lb) patient. The center of gravity changes as the cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to [Chapter 6](#) for gantry and table mounting details.

No part of the floor surface within the table and gantry, nor the two interface areas between the table and gantry, should be higher than the support area for the table and gantry.

5.2 Floor Anchors

Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements. At the customer's expense, all other anchoring methods (on floor types other than the 102 mm (4 in.) concrete minimum) must be determined by their structural contractor to meet the stated GE minimum load requirements. The customer's contractor is responsible for the installation of all anchors other than those shipped with the system.

5.3 Floor Strength

Concrete floors must have a minimum strength of $f'_c = 2000$ psi (1.4×10^7 Pa) at 28 days (curing time) for mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength.

Note: If installing the GE LightSpeed scanner on a floor type thinner than a 102 mm (4 in.) concrete floor, the purchaser, at their expense, shall provide acceptable anchoring and mounting methods that meet all structural specifications provided in sections [Section 5.0](#) of this Pre-Installation Manual.

5.4 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 3 mm (0.125 in.) between depression and high spots over any 3048 mm (120 in.) distance within the area of the gantry/table template (see the envelope shown in [Figure 2-1, on page 41](#)).

The use of floor shims is not suitable to achieve floor levelness. It is recommended that the concrete be leveled to meet this requirement.

The floor must meet levelness specifications to properly align the table and gantry. The minimum gantry height is 15 mm (0.5 in.) to prevent cable crushing.

Table levelness may not be achievable if the overall floor levelness is greater than the specification. The overall floor level must be 0 in. to use the under gantry cable entrance. To use this option, the minimum gantry height must be 20 mm (0.75 in.) to prevent cable crushing.

5.5 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits. However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

5.5.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

5.5.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

5.5.3 Equipment Location

To minimize the interference, place the CT equipment on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Trains
- Heliports
- Roadways
- Hallways
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units
- Subways
- Elevators

5.6 Walls: Scan Window

The recommended patient viewing window dimensions are 1219 mm wide x 1067 mm high (48 in. wide x 42 in. high). The location of the window is dependent on the position of the operator workspace position. Consult [Section 7.0](#) of this chapter and a **qualified radiological health physicist** for radiation protection requirements of the window glass (lead content and thickness).

Note: The operator at the operator workspace must be able to view the patient during a scan.

Section 6.0 Network Connections

The CT system is connected to the network through the console. Use a patch cable (not to exceed 3.1 m (10 ft.)) to connect the console to a wall box. A high speed connection of 1000 baseT with 100 baseT is acceptable.

Broadband is the standard network connection and is required for all LightSpeed systems. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the console.

-
- A patch cable (not to exceed 3 m (10 ft.)) should be provided by the customer to connect the operator console to a wall box. (See Notes on [Figure 9-4](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera, and operator console.
- The run from the hospital switch to the CT wall outlet must not exceed 88 m (290 ft.). Bandwidth performance is degraded when the length reaches 91 m (300 ft.) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice-only line.



Figure 4-15 Console Rear Bulkhead

6.1 US Broadband Process Overview

The US network connectivity requirement for this product is broadband. The US process relies on the GE PMI to select a Customer Champion and identify an IT contact for the site. Together, those

individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

6.2 Customer Broadband Responsibilities

Provide the GE PMI with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Coordinates VPN activities between Radiology/Cardiology and the Information Technology (IT) departments.
 - Acts as a focal point in assuring site broadband infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact: Completes an equipment assessment with the GE Healthcare Connectivity team to determine site readiness for broadband.
- Contact your GE PMI, for the name of the zone broad-band specialist to:
 - Work with the Customer Champion to complete any identified infrastructure changes.
 - Provide an IP addresses for new CT equipment.
 - Provide a VPN compatible appliance that supports the IPSec tunneling protocol and 3DES data encryption.
 - Utilize an Internet Service Provider that supports static routing.

Section 7.0 Radiation Protection

Scanner room shielding requirements should be reviewed by a qualified radiological health physicist taking into consideration equipment placement, weekly projected work-loads, and materials used for construction of walls, floors, ceiling, doors, and windows.

Figure 4-16 depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and Figure 4-17 depicts a 20 cm water phantom (head) with the technique shown.

Note: **Actual measurements can vary.**
All measurements have an accuracy of $\pm 20\%$ because of measurement equipment, technique, and system-to-system variation.

Use the correction factors shown in Table 4-9 to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75 mm images	0.82
16 x 0.625 LD 8 x 1.25 LD Fluro 5mm	0.59

Table 4-9 Shielding Requirements Scaling

CHANGED PARAMETER	MULTIPLICATION FACTOR
4 x 1.25 LD 5mm (1i) Fluro 2.5 mm	0.40
1 x 1.25mm images	0.20
4 x 0.625 LD 1 x 1.25	0.10

Table 4-9 Shielding Requirements Scaling

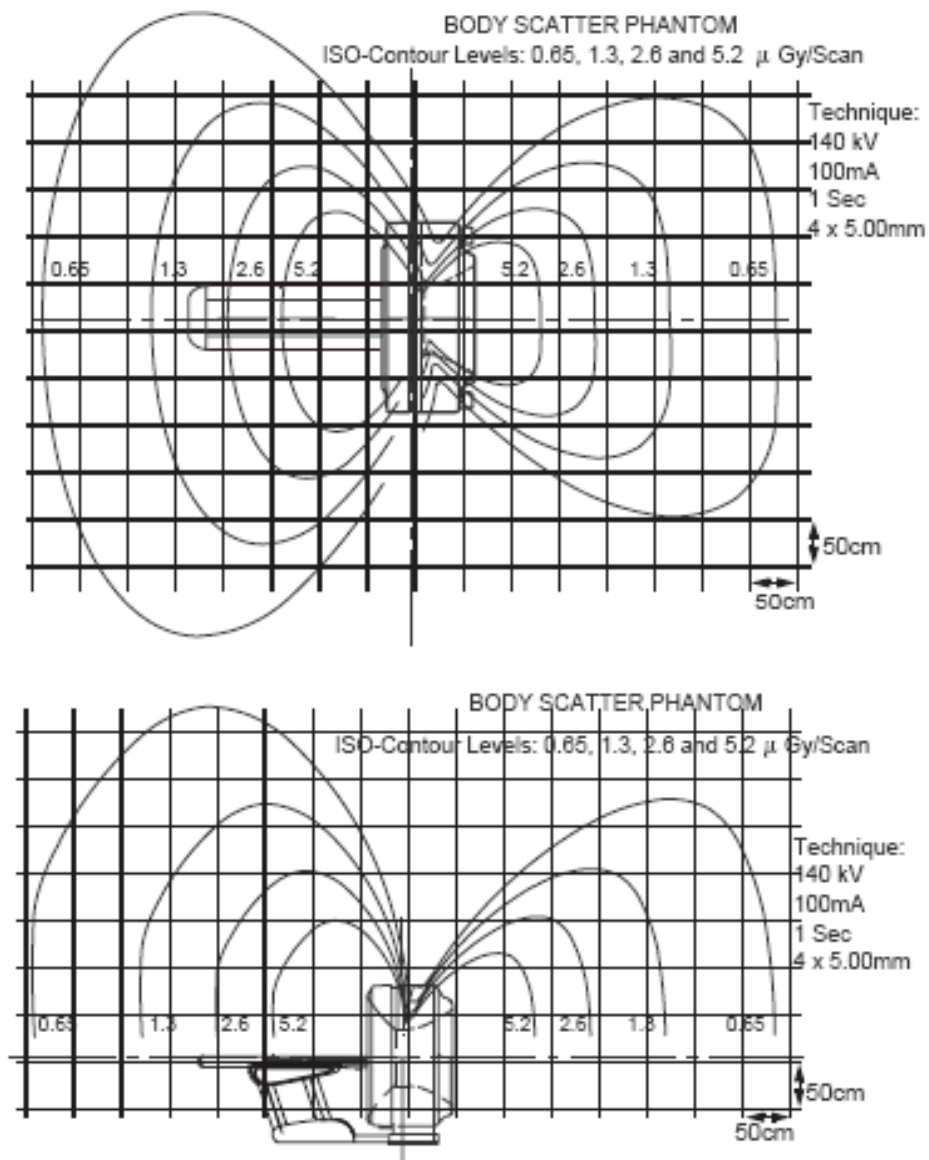


Figure 4-16 Typical Scatter Survey (Body Filter)

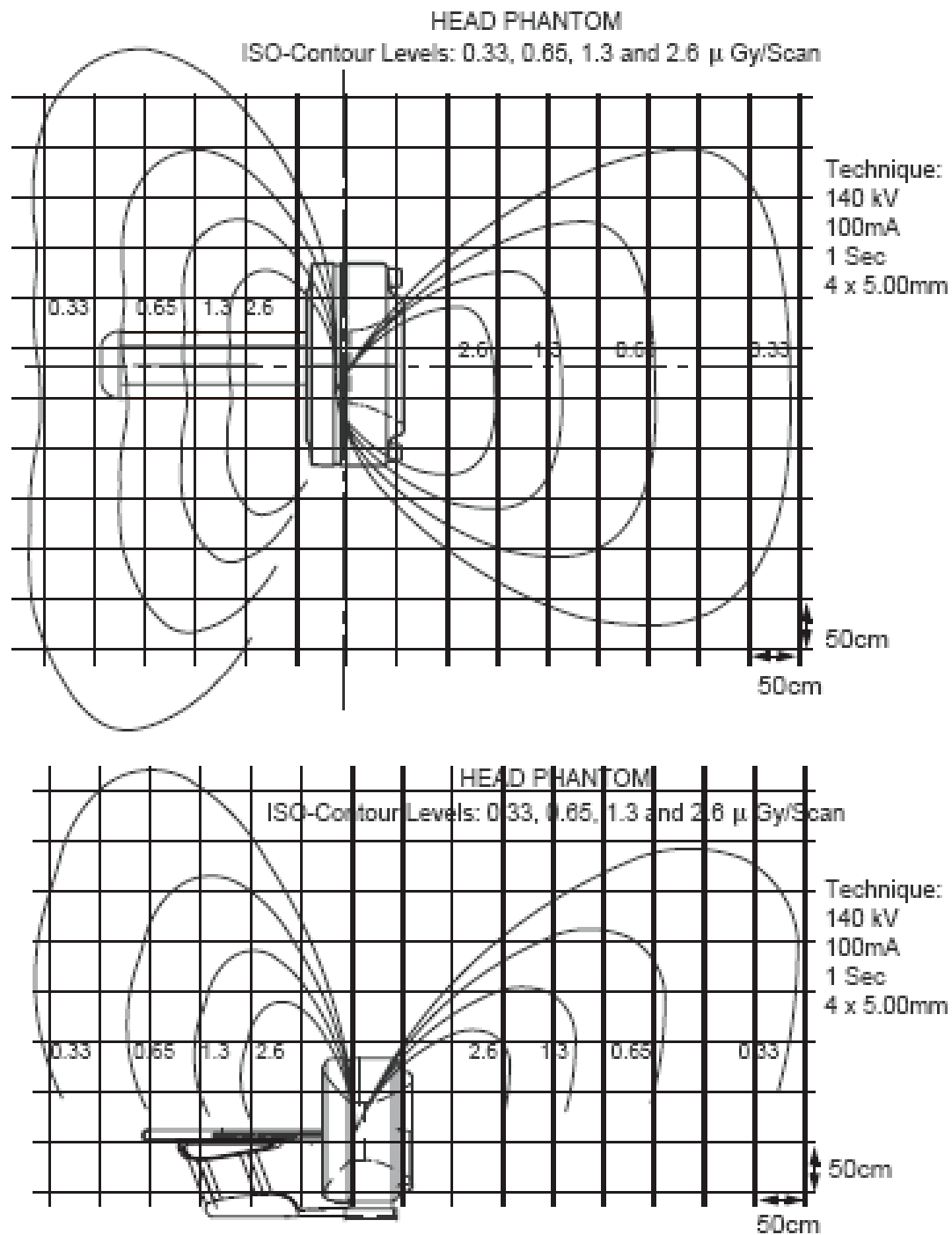


Figure 4-17 Typical Scatter Survey (Head Filter)

Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if the site environment meets the standards of this section. Maintain environmental conditions listed below at all times—including overnight, weekends, and holidays. Shut down the CT system if the air conditioning is not working. When the system is shut down for major repairs, the air conditioning also may be shut down.

Section 1.0

Temperature and Humidity Specifications

- Ambient Temperature: (Fahrenheit and Celsius)
 - Scan Room: Maintain a temperature of 64° - 79° F (18° - 26° C), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 64° - 79° F (18° - 26° C).
 - Control Room (including Console/Computer): Maintain 64° - 79° F (18° - 26° C).
 - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 64° - 79° F (18° - 26° C).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5° F/hr (3° C/hr).
- The maximum relative humidity rate of change is 5% RH/hr.

Section 2.0

Temperature and Humidity Monitoring

Locate computing subsystems in an area that has the environment specified in [Section 1.0](#), above. First, assess your environment's heat and humidity. If necessary, temporarily install a temperature and humidity recorder close to where the gantry will be installed. Note readings before installation and again after installation to verify true temperature and humidity conditions for your environment. Consider your HVAC needs and redundancy. You may wish to consider an air conditioner with two compressor units rather than one. A backup (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 3.0

Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. For patient comfort, locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort.

SYSTEM COMPONENT	ROOM =>	SUITE	
		BTU/HR	WATT
SCANNER SYSTEM			
1. Gantry LS16 ¹		32,100	9400
2. Table		700	20
3. Power distribution unit		3400	1000
4. Operator's console/computer with two monitors		6800	2000
4b. Operator's Console/computer with one IG		7361	2165
Additional IG, each		1360	400
LCD monitor, each (x2)		170 (340)	50 (100)
SCSI Tower		425	125
Injector		425	125
DASM		425	125
Recommended Control Room subtotal (includes OC w/1 IG, 2 monitors & SCSI Tower)		8126	2390
SYSTEM TOTAL (LS 16)		43,000	12,600
SELECTED OPTIONS			
Powerware UPS		2500	732
Liebert UPS		2500	732
Remote Color Monitor		1178	345
Advantage Windows		256	75
ROOM TOTAL²			

¹ With 75 scan rotations per patient, minimum gantry cooling is enough for up to three patients per hour. Recommended cooling is enough for up to four patients per hour. Use growth BTU/hr. to size cooling for future, more powerful X-ray tubes.

² Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-8](#), [Figure 4-11](#) (CPDU), [Figure 4-13](#) (NGPDU), and [Figure 4-14](#), in Chapter 4, [Section 1.0](#), for component air flow requirements.

Section 4.0

Altitude

System operating altitude is from mean sea level to 3050 m (10,000 ft.).

Section 5.0

Electro-Magnetic Interference (EMI)

Note: If power sub-stations exist under or above the scan room, or near the control room, consider EMI testing to determine if your proposed room meets the published acceptable EMI room limits. This includes high voltage lines under the scan or control room floor.

5.1 Gantry

Locate the gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.

5.2 Color Monitor

Locate the color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (500 milligauss) to guarantee color purity and display geometry. See [Figure 5-1](#).

5.3 Console/Computer Equipment

Locate the computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-1](#).

5.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

5.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., operator console or computer equipment—the UPS is not classified as sensitive electronics) within 1.5 m (5 ft.) of the Power Distribution Unit, in any direction (including above or below). See [Figure 5-1](#).

5.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Healthcare Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from the source of the magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single-phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.
- Maintain good screening of cables and cabinets.

5.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least 1 m (3.3 ft.) away from sensitive electronics (the PDU does not include sensitive electronics).

for UPS interconnection information, refer to [Chapter 9, Section 5.0 on page 110](#).

5.8 Equipment EMI “Envelopes”

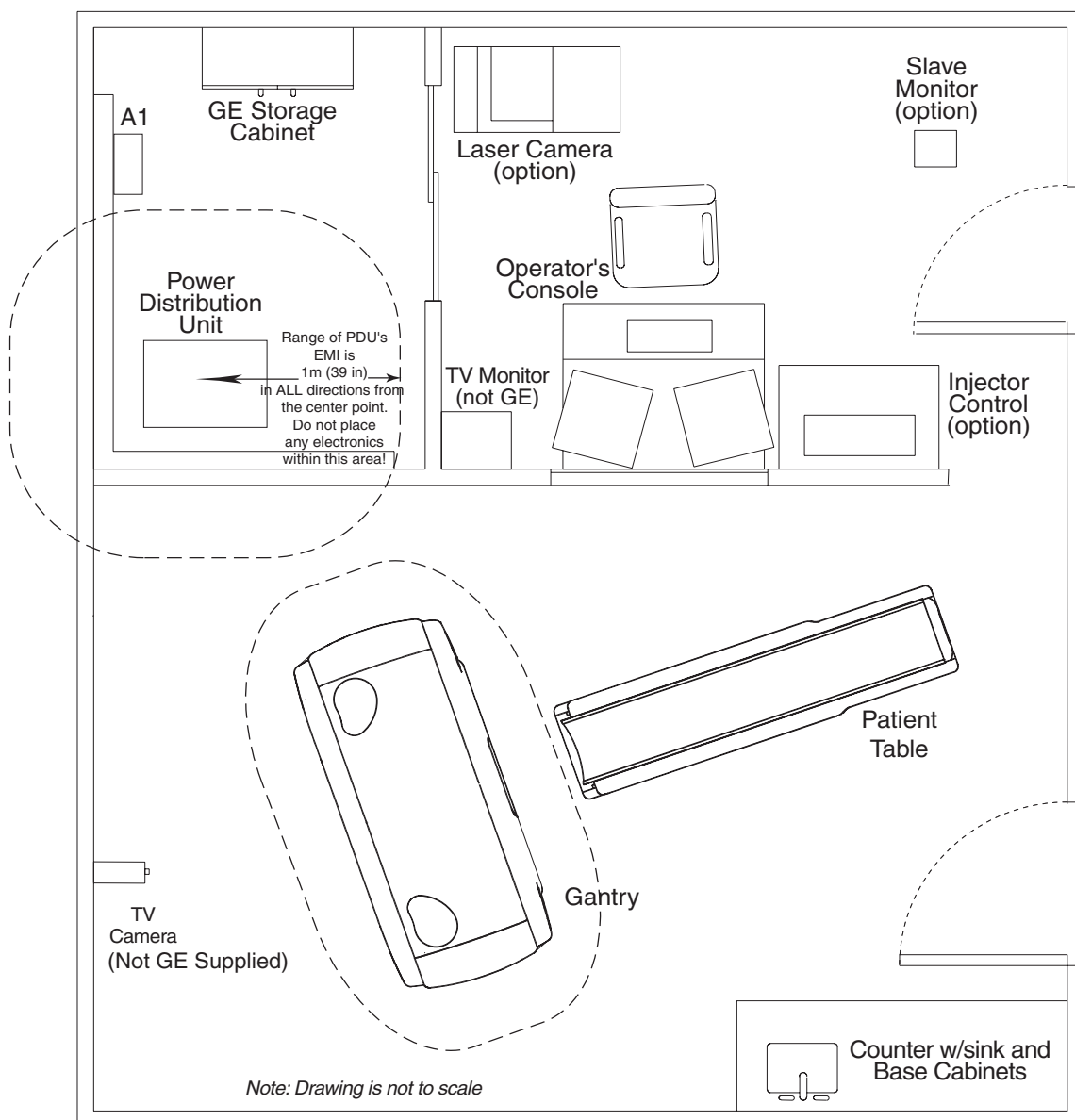


Figure 5-1 Sample Room Layout, showing approximate EMI requirements

5.9 EMC Edition 2 Compliance

5.9.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The LightSpeed system is suitable to use in the electromagnetic environment, as per the limits and recommendations described in your technical reference manual.

Adhering to the distance separation recommended in tables found in the technical reference manual, between 150KHz and 2.5 GHz, reduces disturbances recorded at the image level, but may not eliminate all disturbances. However, when installed and operated as specified herein, the system maintains its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

For example, make sure a 1W mobile phone (800 MHz to 2.5 GHz carrier frequency) is 23 m (75.5 ft.) away from the LightSpeed (in order to avoid image interference risks).

5.9.2 Use Limitation: External components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMAGNETIC COMPATIBILITY of the EQUIPMENT and /or SYSTEM.

5.9.3 Installation Requirements and Environment Control

In order to minimize interference risks, the following requirements apply:

- Cable Shielding and Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Using cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

- This Product Complies with the Radiated Emission as per CISPR11 Group1 Class A Standard Limits

The LightSpeed is intended for use in non-domestic environments, and not directly connected to the Public Mains Network.

The LightSpeed when used in a domestic environment (e.g. doctors' offices) should use a separate AC power distribution panel and line, with an X-ray shielded room, in order to avoid interferences.

- Subsystem and Accessories Power Support Distribution

All components, accessories subsystems, and systems that are electrically connected to the LightSpeed must have all AC power supplied by the same power distribution panel and line.

- Stacked Components and Equipment

The LightSpeed should not be used adjacent to, or stacked with, other equipment. If adjacent or stacked use is necessary, observe the LightSpeed to verify normal operation in the configuration in which it will be used.

- Low Frequency Magnetic Field

In the case of a digital LightSpeed, the gantry (digital detector) shall be 1m (3.3 ft.) from the generator cabinet, and 1 m (3.3 ft.) from the analog (CRT) monitors. These distance specifications minimize the low frequency magnetic field interference risk.

- Static Magnetic Field Limits

In order to avoid interference on the LightSpeed system, static field limits from the surrounding environment are specified:

- Static field is specified as less than <1 Gauss in the Examination room, and in the Control Area.
- Static field is specified as less than <3 Gauss in the Technical Room.

- Electrostatic Discharge Environment and Recommendations

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup. The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

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Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

The LightSpeed system has a total floor load of approximately 3021 kg (6660 lbs). About 2291 kg (5050 lbs), including patient (204 kg (450 lbs)), is concentrated in the table-gantry assembly.

[Table 6-1](#) shows CT components with size and weight, floor loading and normal mounting requirements.

ITEM	NET WEIGHT kg (lb)	OVERALL W X D mm (in.)	WEIGHT/AREA kg/m ² (lb/sq. ft.)	LOAD PATTERN mm (in.)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry	1500 (3300)	2200 X 1000 (88.6 X 39.4)	1188 (244)	Rectangular base plate 610 X 2057 (24 X 81) with four round pads, each 63.5 (2.5) in contact with floor and supporting 707 kg (1560 lbs) center to center.	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Patient Table	538 (1185) Includes 204 (450) Patient	617 X 2387 (24.3 X 94)	852 (175)	Rectangular base 432 X 1460 (17 X 57.5) with five round pads, each 63.5 (2.5) in contact with floor. Worst-case cantilever pull on any bolt is 561 kg (1235 lb).	Hilti Kwik-Bolt II 12.7 mm (1/2 in) diameter by 203 mm (8 in) long per P/N 2106573 at three leveling pads into concrete floor.
Power Distribution Unit (CPDU)	336 (740)	762 X 559 (30 X 22)	788 (161)	Four Casters support area of 762 X 559 (30 X 22).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Power Distribution Unit (NGPDU)	770 (350)	27.6 X 21.7 (700 X 550)	185 (909)	Four Casters support area of 700 X 550 (27.6 X 21.7).	
Operator's Console (w/o monitors)	159 (350)	1219x991 (48 X 39)	281 (58)	Four Casters or Leveling Feet support area of 1168 X 483 (46 X 19).	Casters are for positioning. Set on floor. Four leveling pads may be anchored to floor using angle brackets ² in seismic zones.
Monitor - CRT (ea)	32 (70)				
Monitor - LCD (ea)	9 (20)				

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in [5.1 - Table and Gantry Mounting Requirements](#), on [page 64](#).]
- 2.) CPDU: Angle brackets are included in the OPTIONAL Seismic Kit (R4390JC) only.
NGPDU: Angle brackets are included with the NGPDU itself

Table 6-1 LightSpeed System Floor Loading

Section 2.0

Mounting Data, Including Seismic

 **WARNING POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-4](#)
- Power Distribution Unit: [Figure 6-5](#)
- Operator's Console/Computer: [Figure 6-8](#)

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available for facilities in areas that require one.

The customer is responsible for seismic mounting. Refer to all applicable codes in your area.

GE-provided floor anchors ([Figure 6-1](#)) are designed for use ONLY on concrete floors that meet the concrete floor requirement. A trained contractor must install the supplied floor anchors, and set them to a minimum depth of 76 mm (3 in.) at each anchor point. Any anchors having more than 25 mm (1 in.) of thread showing above the nut when torqued to 75 N-m (55 ft.-lbs.) must have a second anchor installed in the closest adjacent hole. Use standard depth and torque specifications for the second anchor.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum floor thickness	102 mm (4 in.)	102 mm (4 in.)
Recommended drilling depth	95 mm (3.75 in.)	95 mm (3.75 in.)
Average anchor embedment	89 mm (3.5 in.)	89 mm (3.5 in.)
Minimum anchor embedment	76 mm (3 in.)	76 mm (3 in.)
Available alternate anchor locations	Yes	Yes
Shipped anchor size	203 mm (8 in.)	203 mm (8 in.)
Alternate anchoring methods	Yes (see notes, below)	Yes (see notes, below)
Floor levelness requirement	8 mm (0.3125 in.) over 3 m (10 ft.)	8 mm (0.3125 in.) over 3 m (10 ft.)

Table 6-2 Gantry and Table Mounting Requirements

- Note: If the installers cannot set all four anchors for the table, the installer must inform the customer that the minimum anchoring cannot be met. A structural engineering contractor must determine the anchoring method and certify that their anchoring meets the stated GE minimum load requirement and torque specifications.
- Note: All other anchoring methods on floor types other than the concrete minimum. The anchoring method must be certified to meet the stated GE minimum load requirement and torque specifications.
- Note: If installing the GE LightSpeed scanner on a floor type other than a 102 mm (4 in.) concrete floor, all structural specifications in this document must be reviewed and met.

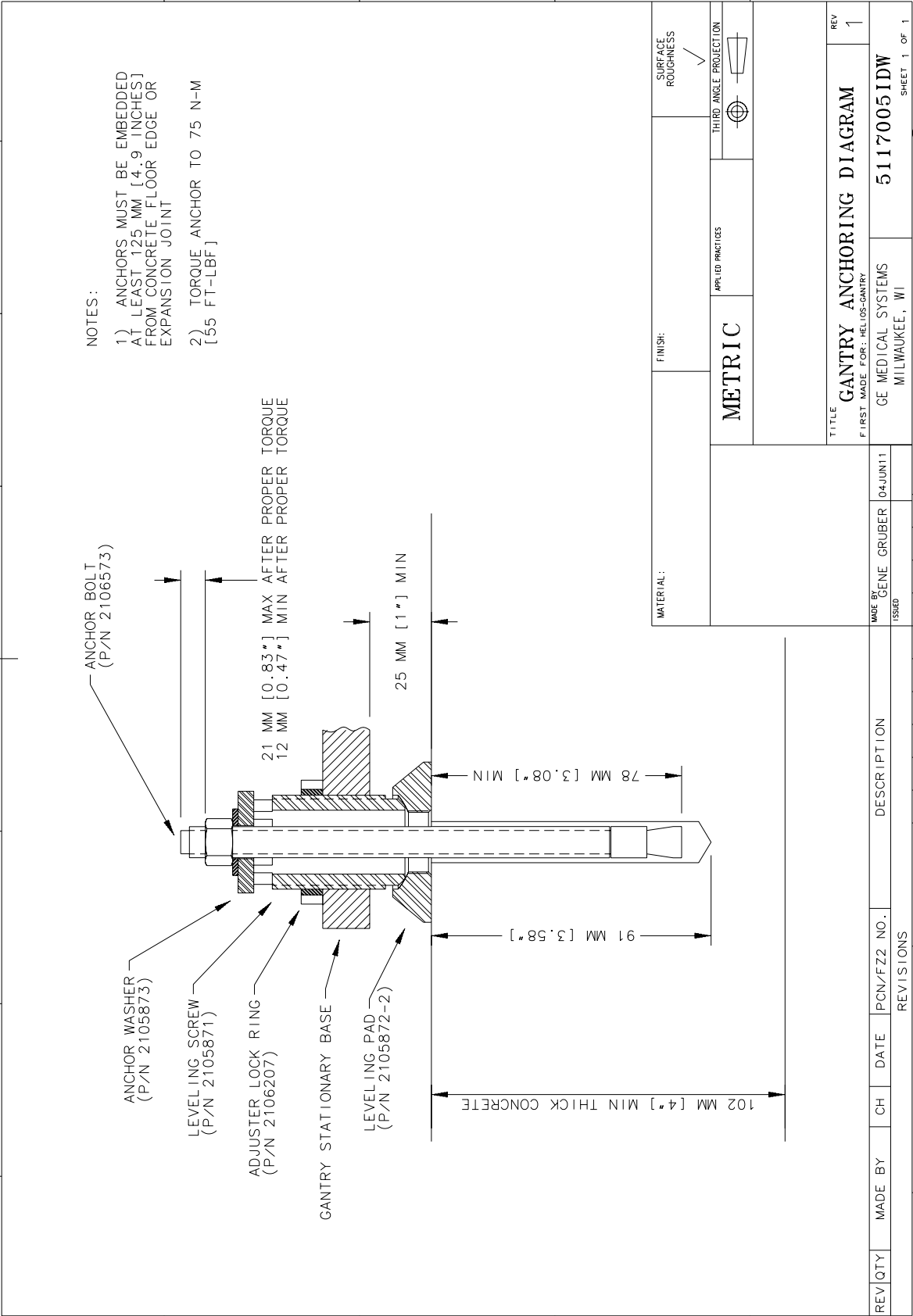


Figure 6-1 Typical Floor Anchor, Gantry and Table

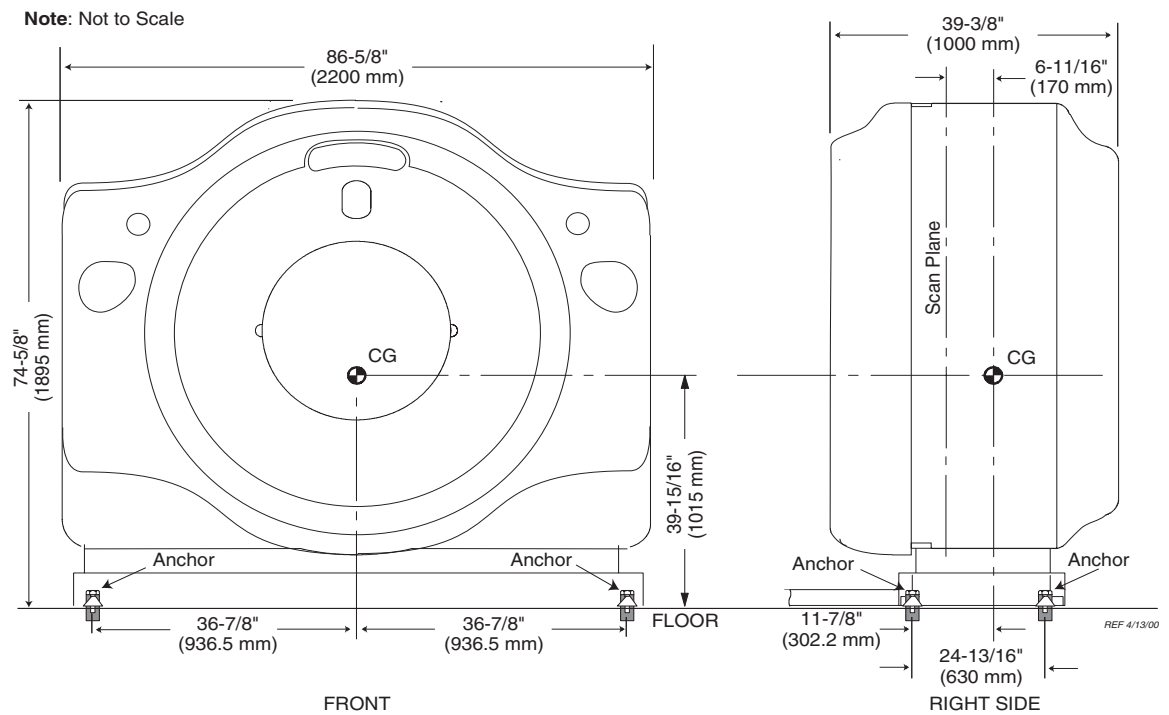
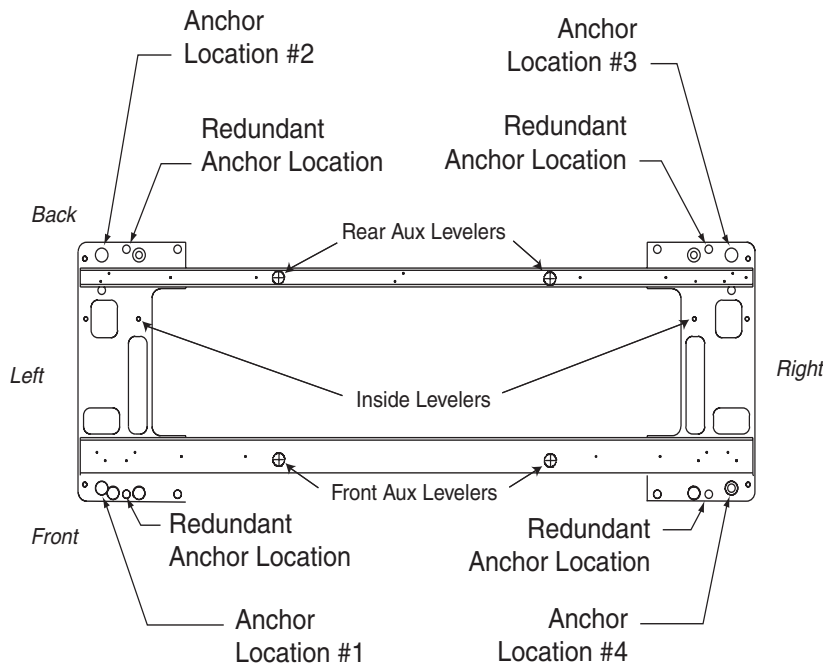


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-3 Gantry Leveler and Adjuster Locations

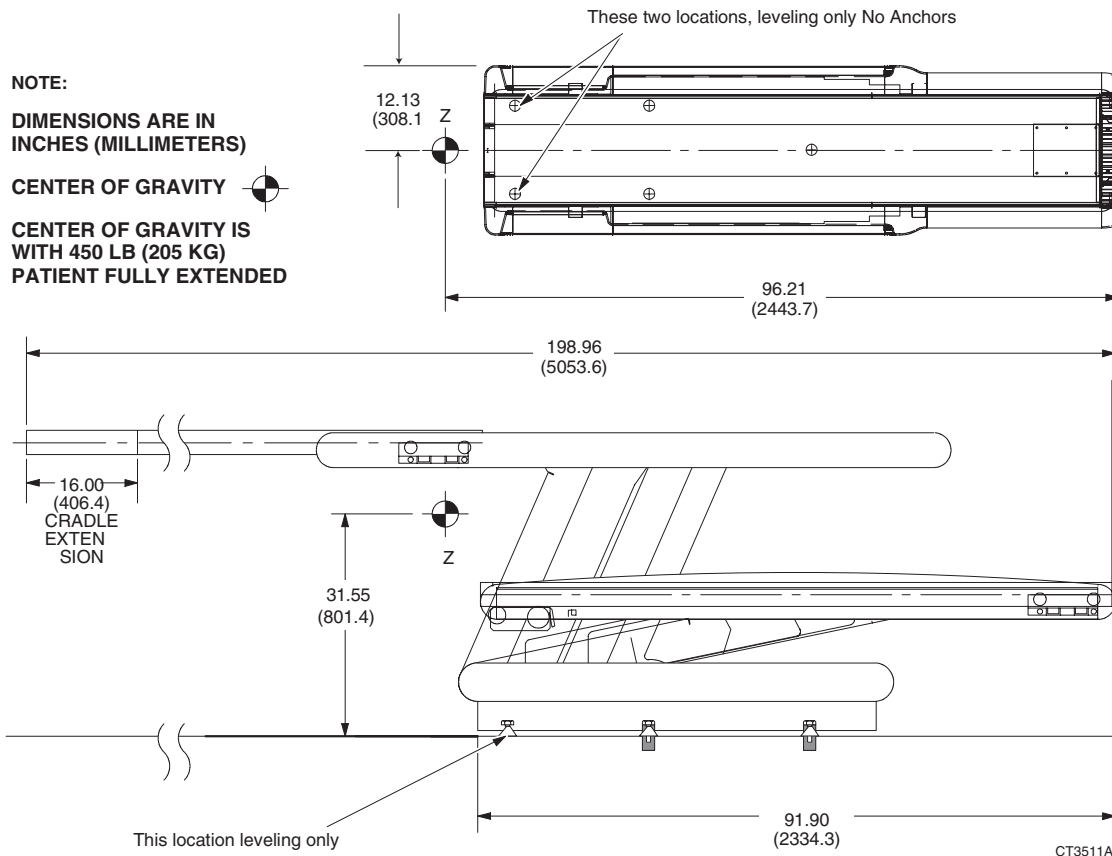


Figure 6-4 Patient Table (CT1)

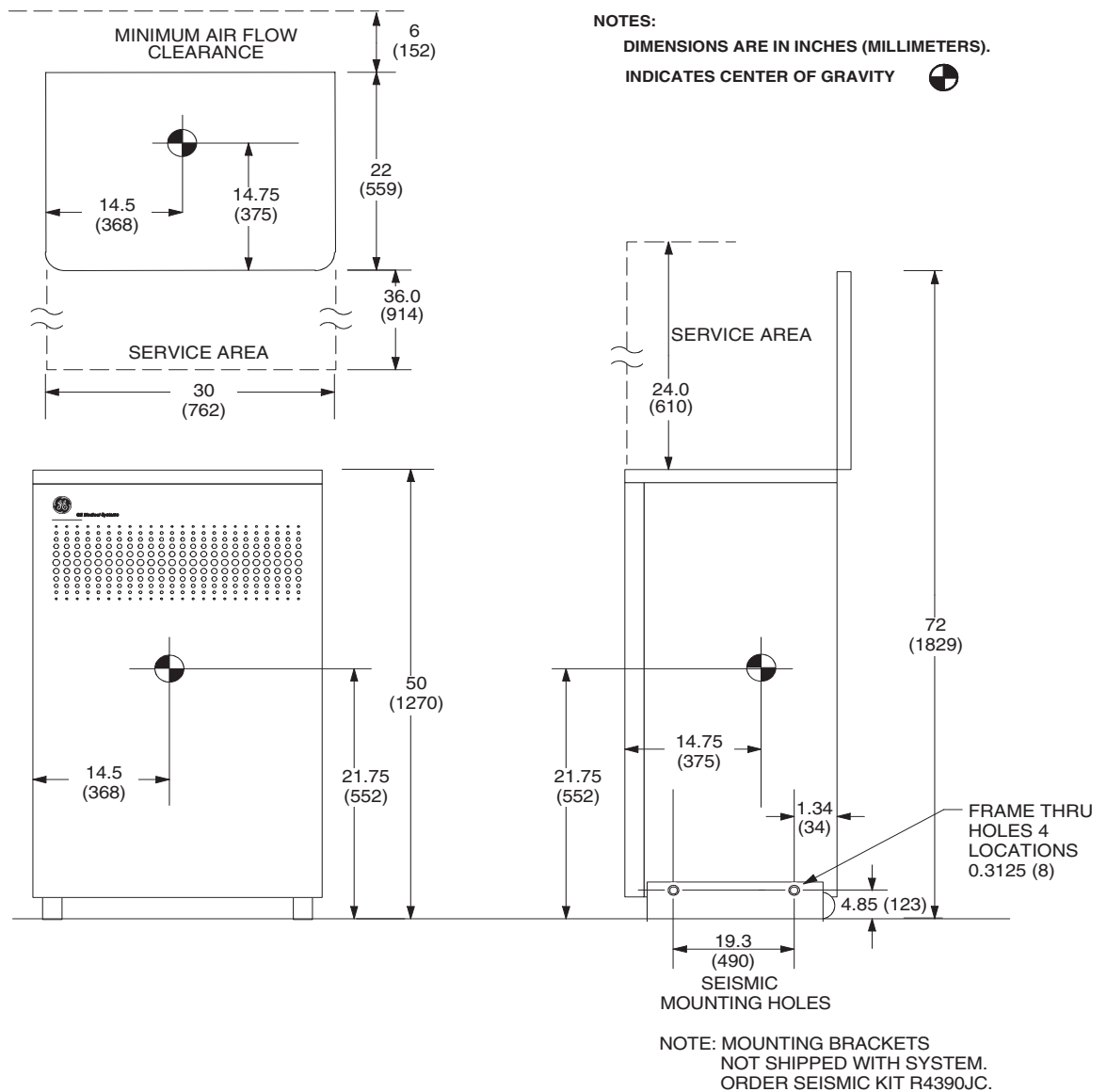


Figure 6-5 Power Distribution Unit (CPDU)

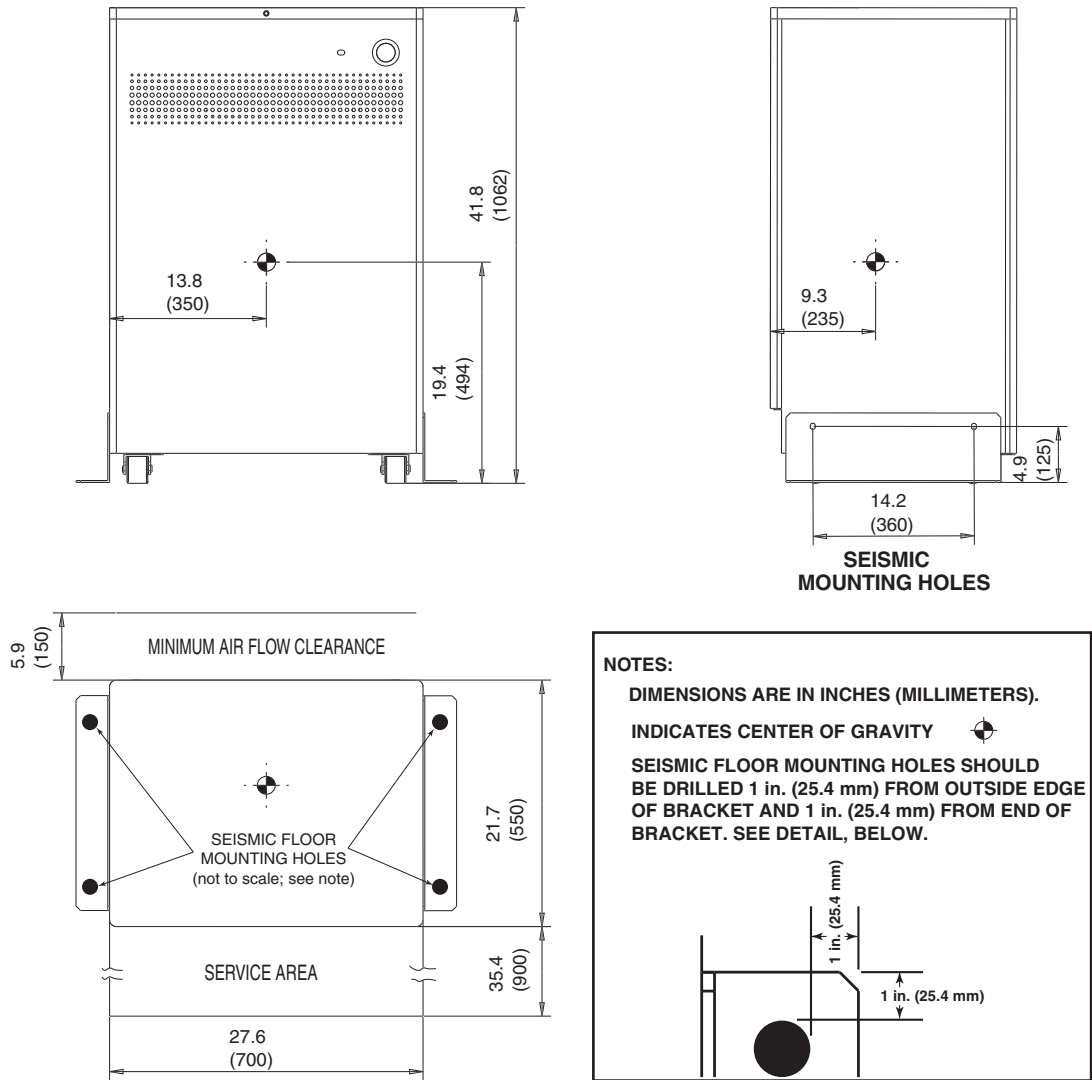


Figure 6-6 Power Distribution Unit (NGPDU)

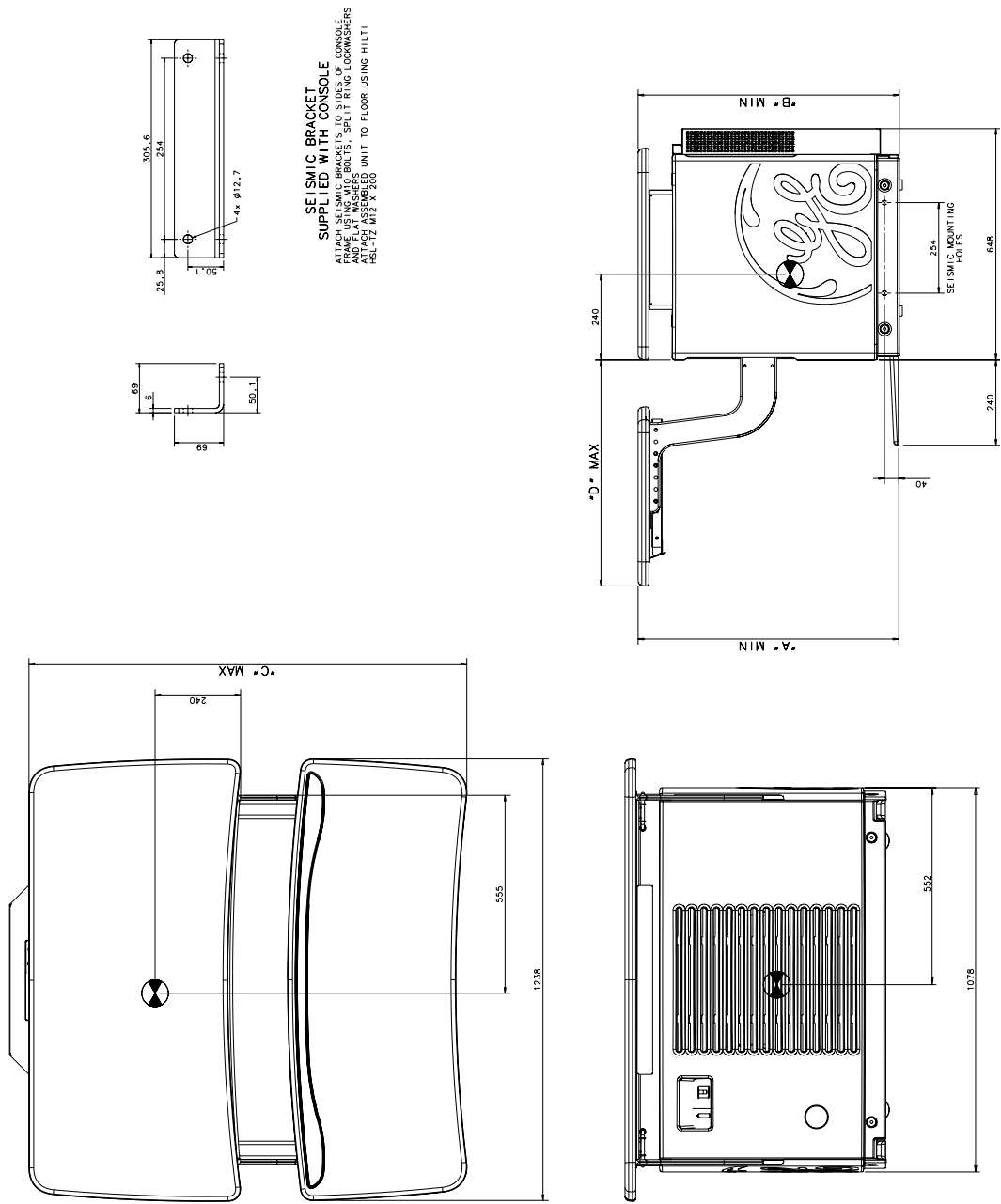


Figure 6-8 Operator's Console (Global Console)



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Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0

Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers which include dollies, skids and boxes without skids.

Section 2.0

Delivery/Shipping Requirements & Considerations

LightSpeed 3.X systems are not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

NEVER drop the gantry, console, table, or PDU. A drop from a height greater than one-half inch ($\frac{1}{2}$ in) may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the gantry, doc-to-dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, moving the system via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces is acceptable.

When moving gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the gantry at the slowest reasonable rate. See [Figure 7-1](#).

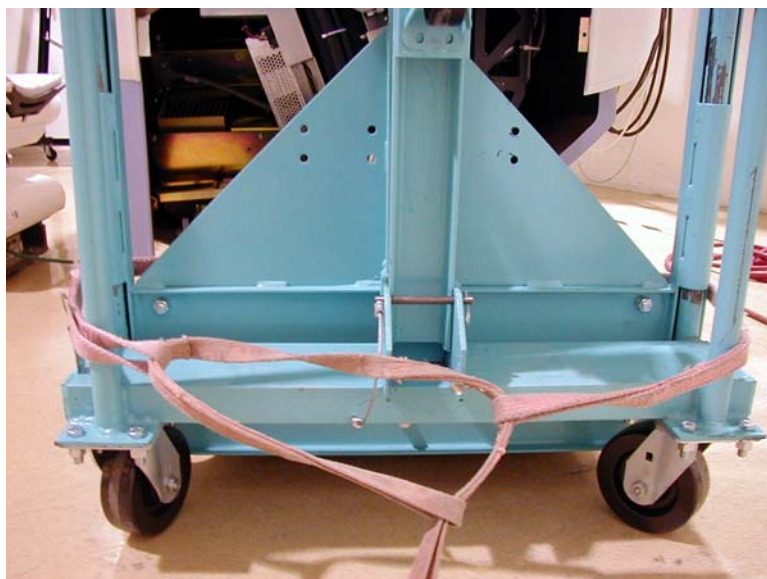


Figure 7-1 Gantry Strap Location

The LightSpeed 3.XSystem—including gantry, console, table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or octane failure) may not be evident until after system installation is complete.

Keep all system components upright at all times; do not tip them. Do not hoist the gantry. Move the gantry only by rolling it on its dollies. Do not tip or lift the gantry when moving it through hallways, doorways, elevators, etc.

Protection for flooring along the move path (from dock to scan room) is advised.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

The LightSpeed 3.Xsystems (consisting of: console, PDU, table and gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, NEVER install these systems in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site can result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Never install wet film processors in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 3.XCT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer’s requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consider the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0 For Crated Delivery

The CT system is packed for air shipment in six (6) packages including the chair. Total weight of the basic system, including pre-installation kit, is 5384 kg (11,869 lbs). It is shipped in wooden crating.

Crate #	Height cm (in)	Width cm (in)	Depth cm (in)	Weight kg (lb)
1	221.5 (87)	145.5 (57.5)	248 (97.75)	2056 (4533)
2	203 (80)	97.5 (38.5)	262 (103)	672 (1481)
3	158 (60)	94 (37)	103 (40.5)	432 (953)
4	130 (51)	146 (57.5)	159.5 (63)	450 (992)
5	157.5 (62)	134 (52.75)	134.5 (53)	298 (656)
6	174 (68.5)	134.5 (53)	134.5 (53)	1476 (3254)

Table 7-1 Typical Crated Delivery Sizes and Weights

Section 5.0 Storage Requirements

Note: Failure to adhere to Storage Requirements will likely result in equipment damage.

5.1 Short Term Storage (Less than 6 months)

If the CT system is being stored before installation, store it in a temperature- and humidity-controlled warehouse. Protect it from weather, dirt, and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 4° to 27° C (40° to 80° F).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum relative humidity rate of change is 5%/hr.

The maximum temperature rate of change is 3° C/hr. (5° F/hr.)

Note: Between delivery is considered short-term storage.
Van storage must meet the same specifications as above.

5.2 Long Term Storage (6 months or more)

If the CT system is being stored before installation, store it in a temperature- and humidity-controlled warehouse. Protect it from weather, dirt, and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperatures should not exceed 10° to 32° C (50° to 90° F).
- Maintain relative humidity (non-condensing) between 20% and 70%.

If you are storing a system longer than six (6) months, but less than 12 months, contact CT Engineering for installation start-up instructions. Storage longer than 12 months is not recommended.

Section 6.0

Extreme Temperature Transportation and Deliveries



NOTICE Failure to adhere to Extreme Temperature Transportation and Delivery requirements will likely result in equipment damage.

Avoid extreme temperatures during system transportation and delivery.

Extreme temperatures are defined as below -18° C (-4° F) or above 49° C (120 ° F), without humidity control

Section 7.0.

System Transportation

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 20% to 80%



NOTICE Component freezing occurs if the CT system is exposed to temperatures below 0° F (-18° C) for more than two days.

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

Section 8.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies (see [Figure 7-2](#)). Two side rails are bolted to the dollies to stabilize the dollies and protect the gantry. Use dolly elevating casters to lift gantry off its base and roll it into position.



Figure 7-2 Gantry with Shipping Dollies and Side Rails

8.1 Door Openings.

Clear door openings for moving equipment into building must be 1067 X 2083 mm (42 X 82 in.) minimum, if there is an 2439 mm (8 ft.) corridor width.

8.2 Elevator requirements.

To move the gantry from the receiving location to the scanning room, consider elevator capacity and size. By removing side rails and one dolly after the gantry is placed in elevator, the gantry width/length and elevator depth requirements are reduced. Contact a representative of the elevator manufacturer if the gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH CM (IN)	WIDTH CM (IN)	HEIGHT CM (IN)
Dollies On, Side Rails On	290 (114)	129 (51)	200 (79)
Dollies On, Side Rails Removed	290 (114)	100 (39.4)	79 (200)

Table 7-2 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door sizes for the gantry with covers and dollies attached, but side rails removed, is 101.6 cm (40 in.).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

8.3 Dollies

Dollies typically are used on the gantry, table, and console for domestic shipments. Once the gantry, table, and console are installed at the site, return the dollies to GE using the shipping document located in Box #1.

Purchase dollies for international shipments (B7850LD) to use at the customer site. After the system is removed from the crates, dollies with international shipments **only** are not to be shipped to GE in Milwaukee, WI, USA, but returned to the local GE office or warehouse.

Zero-clearance dollies are available. For information on zero-clearance dollies, please go to: <http://www.umi-dollyshop.com/default1.htm> and order.

Section 9.0 Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 2565 mm long x 81 mm wide x 110 mm high (101 in. long x 32 in. wide x 47 in. high).

Section 10.0 Console Considerations

Although not installed, the keyboard table is shipped with the console.

Console shipping dimensions are 1016 mm deep x 1346 mm wide x 1041 mm high (40 in. deep x 53 in. wide x 41 in. high).



NOTICE Do not lift the console by the monitor table top to remove it from the shipping skid.



Figure 7-3 Console, ready to be unloaded from shipping skid

Chapter 8

Power Requirements

Section 1.0

Introduction

The power distribution unit (PDU) supplied with LightSpeed systems transforms and distributes power to all system components. The PDU is the only power entry point required to operate the system.

Keep the power wiring between the facility main distribution panel and the PDU as short as possible. This minimizes voltage regulation effects.

Carefully consider advantages and disadvantages of conduits, floor ducts and surface raceways for running cables. Make cable passageways large enough to install any cable with all other cables already installed.

Excess cable may be stored in conduits, a cable storage box, or a floor duct, provided sufficient space is available. Observe the fill rate with each option. If you have questions regarding local electrical or building codes, consult the project's electrical contractor or electrician.

When routing power cables, all three-phase wires and ground must be run in the same conduit or raceway duct-work. Route power cables separately from system control cables (for example, use a separate trough in the duct).

Section 2.0

System Input Power

2.1 Facility Source

Supply power to the system by a dedicated feeder from the nearest Main Distribution Panel (MDP). Provide a protective disconnect device in the power line supplying the PDU in accordance with National Electric Code and applicable local codes.

Note:
Lockout/tagout
provision
required

Locate the disconnect device within forty feet of the PDU, visible to PDU service personnel, and have a provision for tagout / lockout. It is identified as "A1" in the interconnection schematic diagrams. (GE part number for the disconnect is R4502AD.)

The rating of the disconnect device depends on the nominal line voltage. It must provide over-current protection and have a low voltage release, with multi-point remote control capability. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for minimum rated capacity requirements and suggested devices.

2.2 Main Disconnect Control

Mount customer-supplied Emergency Off buttons in the CT Room near each exit 1424 mm (60 in.) from the floor, and connected to the protective disconnect device, in order to disable the power to all CT system equipment in emergency situations. Clearly label the buttons “Emergency Off” and make them visible to personnel in the CT Room. It is important to label the button “Off” and not “Stop,” since there is an “Emergency Stop” button in the LightSpeed system that disables output power to the CT system equipment in the patient area. Mount an additional emergency button in the computer/equipment room near the exit door.

The customer-supplied main disconnect controller (MDC) must be lockable to meet OSHA standards for power Lockout/Tagout requirements. An optional MDC is available from GE, that provides for the disconnect of the facility power to the PDU, including the emergency buttons, and contacts for an interlock to the air-conditioning units in the computer/equipment room.

The Main Disconnect Control Panel must be UL (or other nationally-recognized testing organization listed) and labeled in accordance with 1999 National Electrical Code (NEC) Article 110-2.

2.3 Configuration

The LightSpeed system is designed to operate on three-phase, **four-wire** wye power. A ground-referenced wye source produces the lowest leakage currents and is preferred. However, the neutral wire does not need to be run to the system, i.e., four-wire connection. (A dummy terminal is provided for ‘parking’ the neutral wire in the event a five-wire service is already installed at the site.)

2.4 Rating

Catalog Number: B7850NP (CPDU)

Voltage 380 to 480 VAC (see note, below)

Capacity 90 calf @ 85% P.F.

Frequency 50 or 60 Hz $\pm$ 1.0 Hz

Catalog Number: B78752AB (NGPDU)

Voltage 200 to 240 VAC, 380 to 480 VAC (see note, below)

Capacity 90 KVA momentary
20 KVA average

Frequency 50 or 60 Hz (47 to 53 or 57 to 63 Hz)

Note: The absolute range of line voltage at the input to the PDU must remain within one of the ranges shown in [Table 8-1](#) at all times.

2.5 Regulation

The size of the facility transformer and feeder wires determine load regulation presented to the system. Total load regulation, as measured at the PDU input terminals, must not exceed 6%.

2.6 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.7 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in [Table 8-1](#).
Limit maximum transient voltage to 1500V peak.

2.8 Microcuts

LightSpeed systems generally are unaffected by microcuts.

2.9 Grounding

Originate the ground to the CT system at the system power source, (i.e., transformer or first access point of power into a facility), and be continuous to the CT system power disconnect in the room.

Run a dedicated 1/0 (55 mm²) or larger insulated copper ground wire with the phase wires from the main distribution panel to the PDU. These grounds can be spliced with “High Compression Fittings” and terminated at each distribution panel it passes through. When the ground is broken for a connection to a panel, connect it into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block. Then connect the grounding wire to the steel panel. Do not use the steel, or other material of the panel, as the block.

The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and **earth must not exceed 2 ohms.** Resistance between any two grounded devices must not exceed 0.1 Ohm to ensure equal potential ground system within the CT Room.

Section 3.0

Recommended Power Distribution System

A single-unit installation, where the distribution transformer and feeder are dedicated to the LightSpeed system is recommended. In this case, the minimum recommended transformer size is 112.5 KVA, rated 3.2% regulation at unity power factor. The minimum recommended feeder size and over-current protection device ampacity based on line voltage is shown in [Table 8-1](#).

- Maximum power demand = 90 kVa @ 0.85 PF at a selected technic of 140 kV, 380 mA.
- Average (continuous) power demand at maximum duty cycle = 20 kVa.
- Maximum allowable total source regulation = 6%.

CATALOG >	B7850NP (LIGHTSPEED PDU)								
Nominal Line Voltage (VAC)	200	220	240	380	400	420	440	460	480
Voltage Range (VAC) +/- 8%	184-216	202-238	221-259	350-410	368-432	386-454	405-475	423-497	442-518
Average Line Current (A)	58	52	48	30	29	27	26	25	24
Momentary Line Current (A) @ Nominal Line Voltage	260	236	217	137	130	124	118	113	108
Maximum Line Current (A) @ Low Line Voltage	282	257	235	149	141	135	128	123	117
Primary Disconnect Device (A1) Over-current Protection (A) ⁶	150	150	150	110	110	100	100	90	90
Feeder Length (MDP to A1) ¹									
0-100 ft (0-30.5 m)	3/0 (85) ³	2/0 (70) ³	1/0 (55) ³	2 (35) ³	2 (35) ³	3 (30) ³	3 (30) ³	3 (30) ³	3 (30) ³
100-250 ft (30.5-76 m)	7/0 (215) ³	6/0 (170) ³	5/0 (125) ³	1 (45) ³	2 (35) ³	2 (35) ³	3 (30) ³	3 (30) ³	3 (30) ³
251-300 ft (77-91 m)	8/0 (275) ³	7/0 (215) ³	6/0 (170) ³	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³	1 (45) ³	1 (45) ³	2 (35) ³
301-350 ft (92-106 m)	8/0 (275) ³	7/0 (215) ³	7/0 (215) ³	3/0 (85) ³	2/0 (70) ³	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³	1 (45) ³
351-400 ft (107-122 m)	9/0 (345) ³	8/0 (275) ³	7/0 (215) ³	3/0 (85) ³	3/0 (85) ³	2/0 (70) ³	2/0 (70) ³	1/0 (55) ³	1/0 (55) ³
Sub-Feeder Length (A1 to PDU) 0-40 ft (0-12 m)	1/0 (55) ³	1/0 (55) ³	1/0 (55) ³	1 (45) ²	2 (35) ²	2 (35) ²	3 (30) ²	3 (30) ²	3 (30) ²

Notes:

- 1.) The feeder table above is based on the use of copper wire, rated 75C and run in steel conduit. Wire size is in AWG (mm²). Ampacity is determined in accordance with the National Electric Code (NFPA 70), Table 310-16 (1999).
- 2.) The wire size shown is the minimum allowable for the specified over-current protection rating.
- 3.) The feeder is sized to contribute 2.4% maximum regulation.
- 4.) A minimum 1/0 (55 mm²) ground wire is required.
- 5.) A neutral wire is not required and, if present, is not used.
- 6.) The Primary Disconnect (A1) must have provision for tagout/lockout.

Table 8-1 Facility Power Requirements

If the LightSpeed system must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment which use rapid film changers should be avoided. These changers use a large number of high powered, closely spaced exposures that may coincide with a CT scan and produce image artifacts.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to LightSpeed 4.Xsystem meets all the requirements.

Power Systems for X-Ray Rooms (Direction 46-013833) is available for additional information on power requirements for X-ray systems.

Section 4.0

Uninterruptable Power Supplies (UPS)

Uninterruptable Power Supplies (UPS) are recommended for areas or sites with known power issues. Consult your local power provider for power quality data in your area. UPS is standard equipment on all mobile units. Filter and surge protectors are not needed with LightSpeed systems.

Section 5.0

Power Audit

A site power audit is required for the LightSpeed family of productions site power audit can be arranged with the GE Power Quality team, or through your sales person.

A sample audit form is located in the back of [Chapter 2](#).

Section 6.0 Ground System

The LightSpeed system has been designed to use an equal potential grounding system. The required ground system is shown in [Figure 8-1](#). There are three primary grounding points:

- A system power ground point located in the PM (Power Module).
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in patient vicinity are grounded to the reference ground point.

For additional information, refer to *Electrical Safety Equipment* (Direction 46-014505).

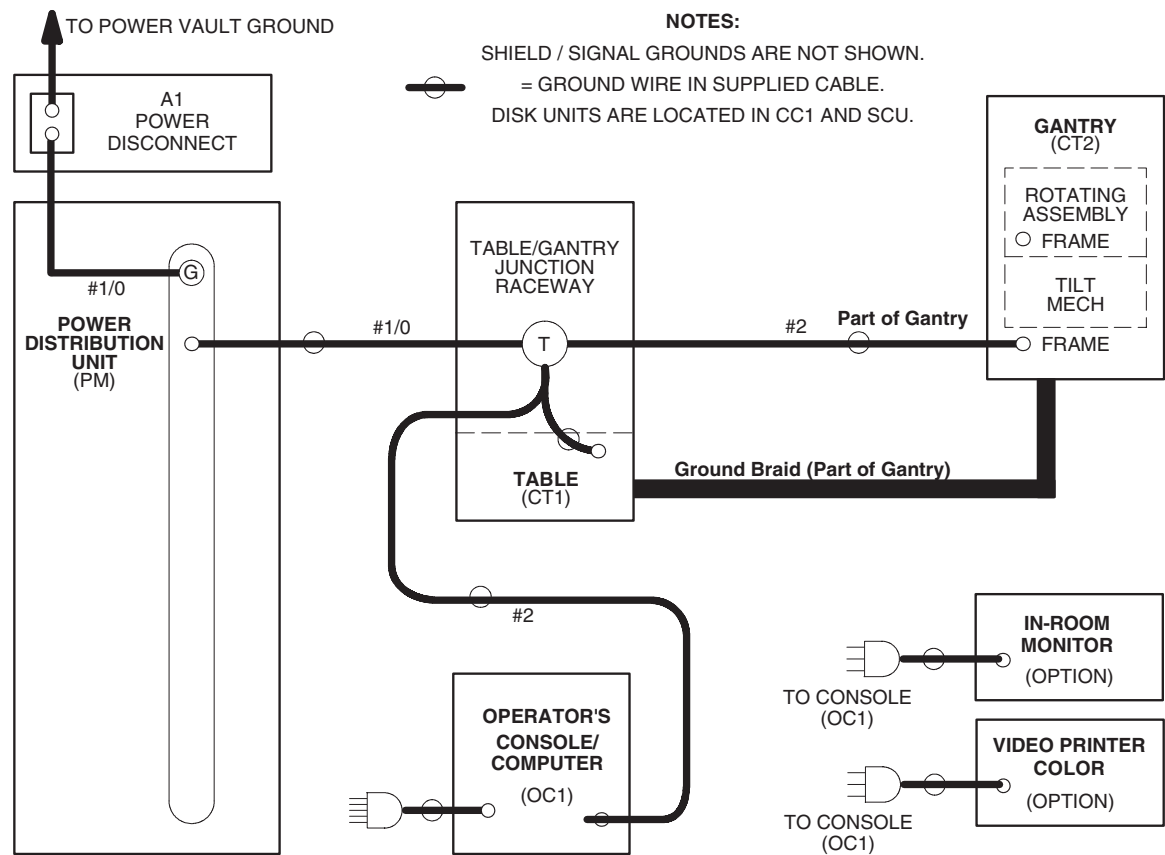


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-4 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-4 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed equipment, using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

The need for additional junction boxes is minimized by using either a cable raceway system or a raised computer floor. LightSpeed systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
LP	Line printer	Option
OC1	Operator's console/computer	System
PM	Power distribution unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
XCVR	Ethernet transceiver	System

Table 9-1 Component Designators

Section 3.0

Interconnect Runs, Wiring and Cables

3.1 GE Healthcare Supplied

3.1.1 Short Cables

Run #	Length, Actual (Usable)		Part NO	Description	UL Styles	Flammability Ratings	Voltage Rating	Voltage (Actual)	Temp Rating (C)	Dia m (in)	# of Cond	Size AWG	Pull Size mm (in)
	ft	m											
001	60 (55)	18.3 (16.76)	2266879-2	Power - PDU to Console	2587	FT-4	600	208Y/120	90	12.3 (0.483)	4	10	56.4 (2.22) Dia
002	43 (35)	13.2 (10.67)	2275844-3	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (0.608)	1	1/0	15.8 (.62) Dia
003	20 (12)	6.1 (3.66)	2267642-2	HVDC - PDU to Gantry	2587	FT-4	600	+ & - 350VDC	90	19 (0.751)	2	6	19.8 (.78) Dia
004	20 (15)	6.1 (4.57)	2266880-2	Power - PDU to Gantry	2587	FT-4	600	208Y/120	90	13.8 (0.542)	5	8	56.4 (2.22) Dia
005	20 (13)	6.1 (3.96)	2267644-2	HVAC - PDU to Gantry	Flexible Motor Supply Cable	FT-4 TC	1000	440Y/254	90	15.3 (0.604)	4	12	16.8 (.66) Dia
006	60 (52)	18.3 (15.85)	2275844-4	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (0.467)	1	2	12.2 (.48) Dia
100	20 (12)	6.1 (3.66)	2266014-2	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (0.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	60 (52)	18.3 (15.85)	2271060-2	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (0.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	60 (55)	18.3 (16.76)	2266887-3	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (0.234)	1	19	15 (.59) Dia
103	60 (55)	18.3 (16.76)	2117848-6	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

Table 9-2 GEHC Supplied Cables (Short Run) - UL Information

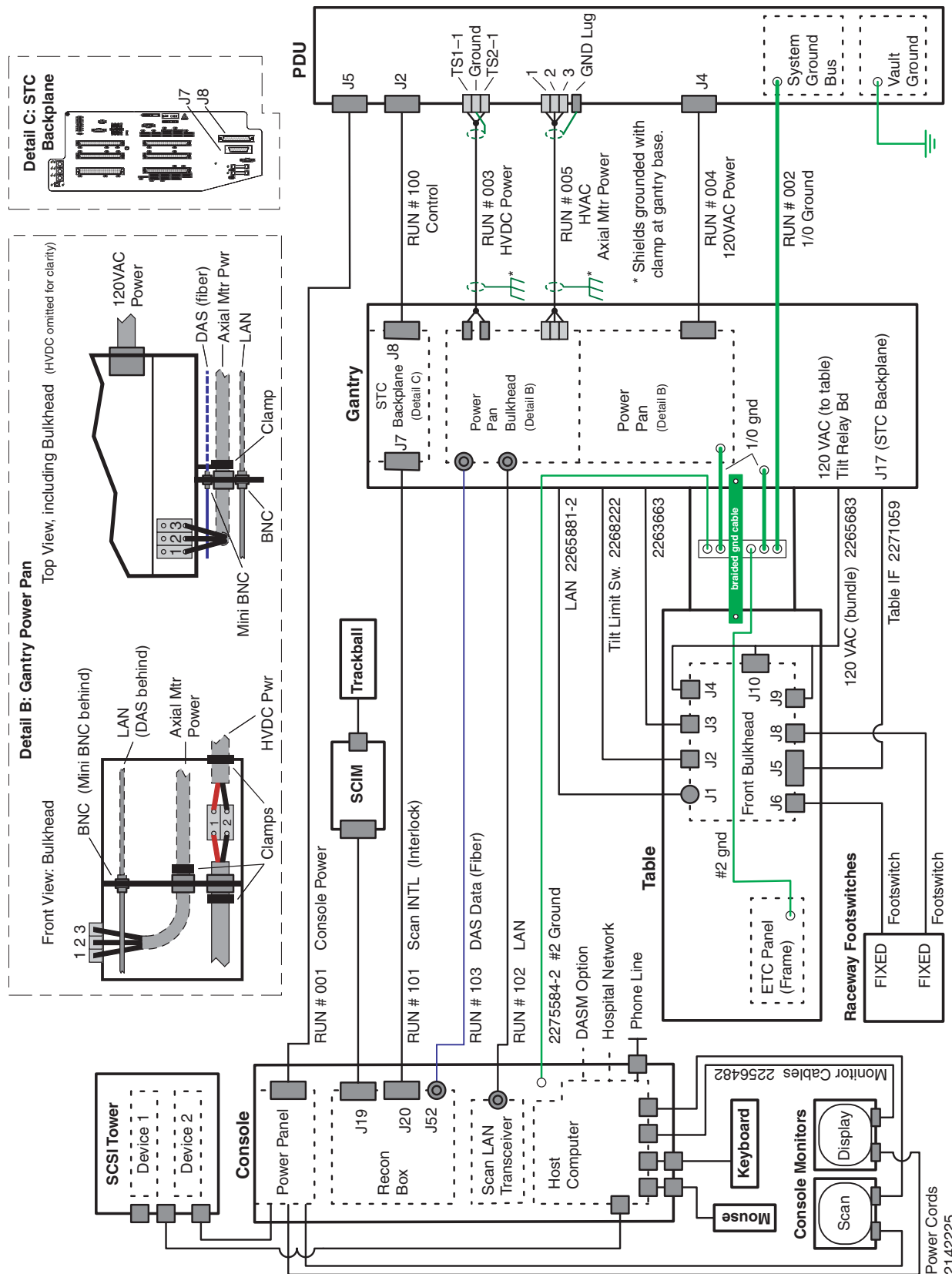


Figure 9-1 System Interconnect Diagram for GOC1 or GOC2 with CPDU

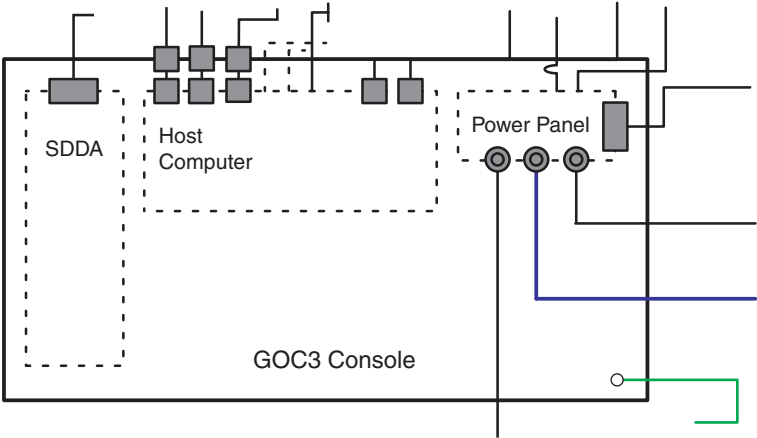


Figure 9-2 GOC3 Console Interconnect

3.1.2 Long Cables

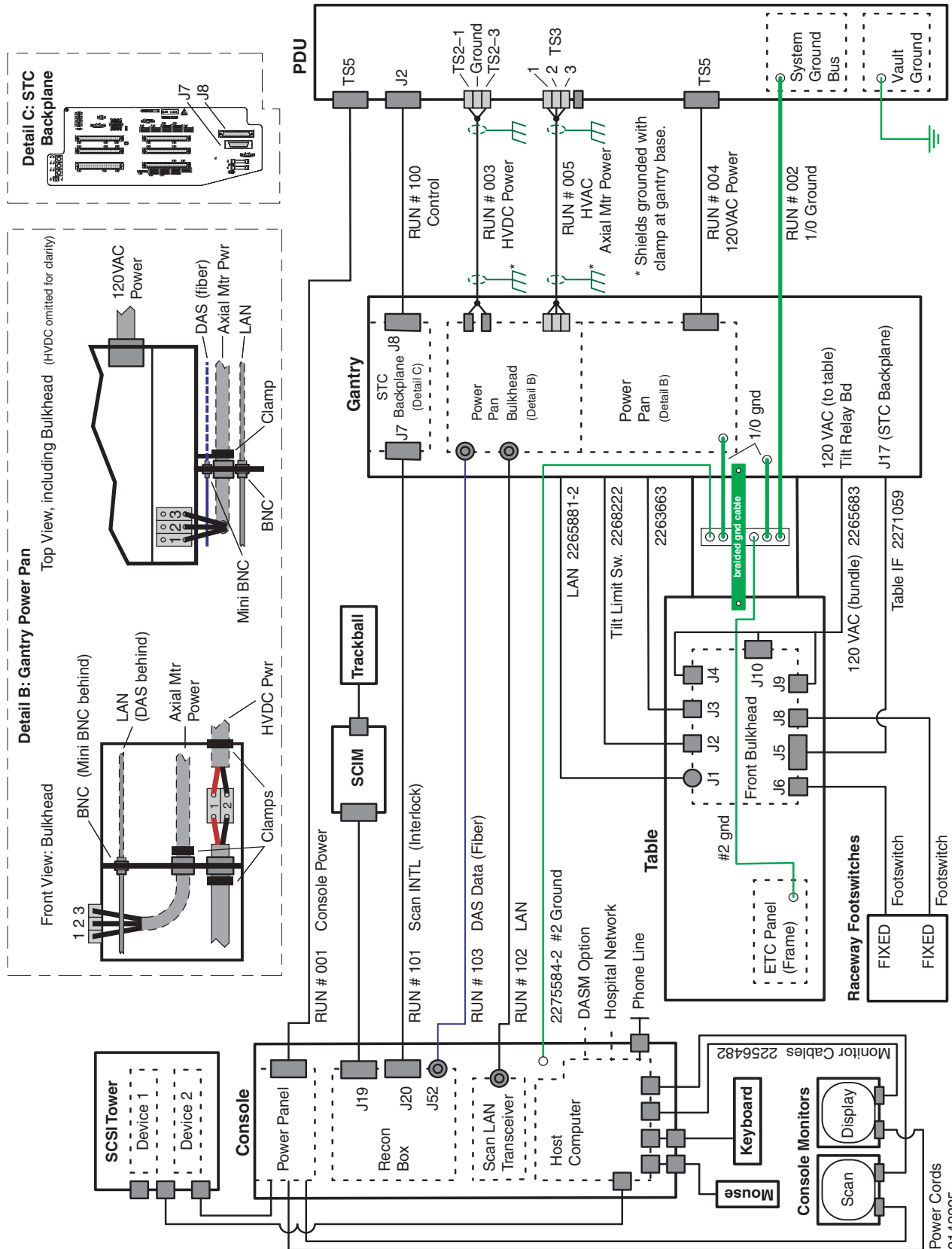
# RUN	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flammability	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
001	80 (75)	24.5 (22.86)	2335376* 2347378 ^N	Power - PDU to Console	2587 2586 ^N	FT-4	600	208Y/120	90 105 ^N	12.3 (.483)	4	10	56.4 (2.22) Dia
002	63 (55)	19.3 (16.76)	2275844	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
003	63 (55)	19.3 (16.76)	2267642 2347513 ^N	HVDC - PDU to Gantry	2587 2586 ^N	FT-4	600	+ & - 350VDC	90 105 ^N	19 (.751)	2	6	19.8 (.78) Dia
004	60 (55)	18.5 (16.76)	2266880 2347377 ^N	Power - PDU to Gantry	2587 2586 ^N	FT-4	600	208Y/120	90 105 ^N	13.8 (.542)	5	8	56.4 (2.22) Dia
005	62.5 (55)	19 (16.76)	2267644 2347514 ^N	HVAC - PDU to Gantry	Flexible Motor Supply Cable 2586 ^N	FT-4 TC	1000 600 ^N	440Y/254	90 105 ^N	15.3 (.604)	4 3 ^N	12	16.8 (.66) Dia
006	83 (75)	25.5 (22.86)	2275844-2	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2266014	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	83 (75)	25.5 (22.86)	2271060	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2266887	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

* Shipped with Global Console

** Shipped with Cable Collector

^N For NGPDU

Table 9-3 GEHC Supplied Cables (Long Run) - UL Information



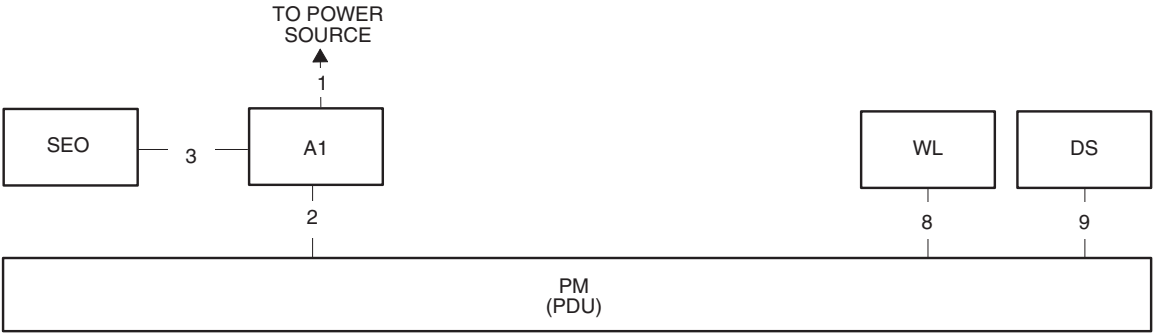
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Figure 9-3 System Interconnect Diagram for GOC1 or GOC2 and NGPDU

3.2 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED		PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M)		
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3 (1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3 (1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 8 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 9 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*		REFER TO FOR AWG (MM2) WIRE SIZES.							

Table 9-4 Runs 1, 2, 3, 8 and 9 Connections



Only one phone connection is required for the system.

- NOTES:**
- 1) Used for remote diagnostics - Option
 - 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
 - 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

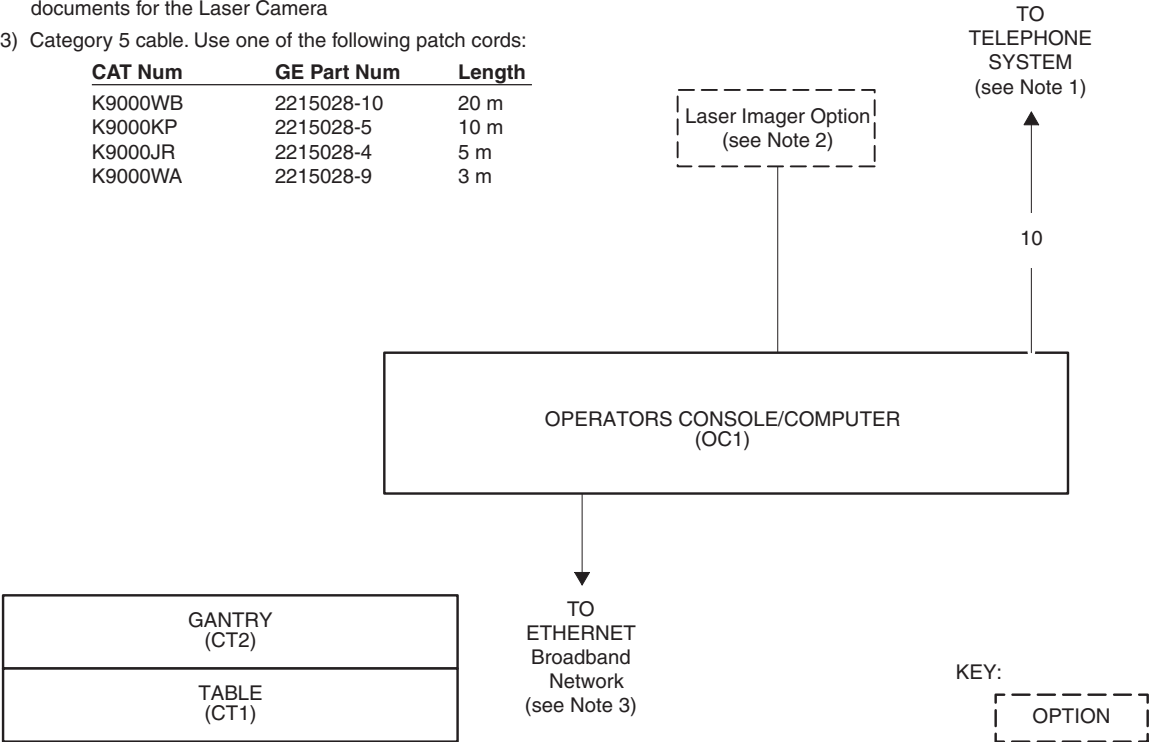


Figure 9-4 Interconnection Runs

Figure 9-5

Section 4.0

Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1 (60 Hz)	Main Disconnect Control (MDC)	480VAC, Surface or Flush Mount, On/Off Control	R4502AD - Main Disconnect Control, 480VAC, Surface Mount with Flush Mount Kit included, two remote Push Button Switches.
A1 (50 Hz) (380V)	Fusible Disconnect and Magnetic Contactor	3 Pole, 380V, Magnetic Combination Contactor. With control trans former for 220V 50 Hz contactor coil, momentary contact start-stop push-button station, indicator light, multiple neutral block, multiple ground block and 160 amp, dual element fuses.	Source Locally
ITL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
MFC	GE LaserCam HQ System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	

Table 9-5 Contractor-Supplied Components

Section 5.0

UPS Interconnect



Figure 9-6 Typical UPS

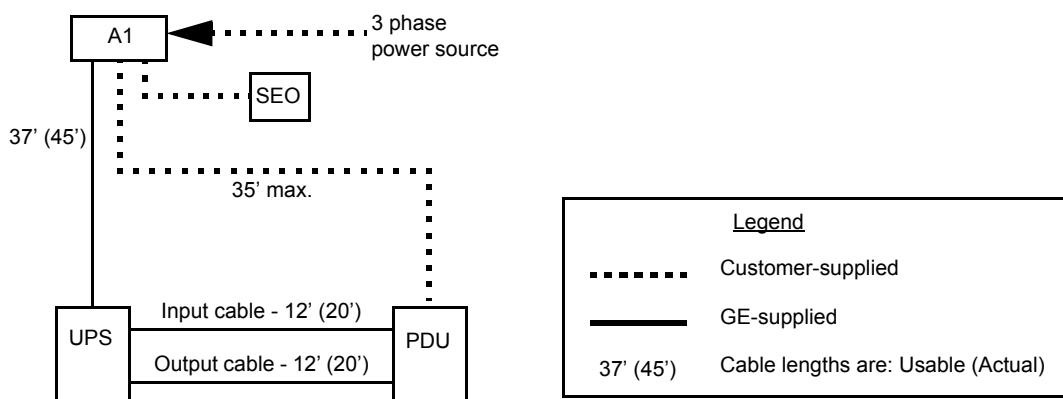


Figure 9-7 Typical UPS Interconnect

PDU Type & Model #	Max. Mom. kVA Rating	Required Main Disconnect (A1) Cat #		Optional Partial UPS Kit Cat #
		Europe & Asia (380-400V or 420V)	North America (440V or 460-480V)	
NGPDU 2326492	90 kVA	E4502AF (150A) (incl. Auto Restart & Integrated UPS Control)	E4502AD (125A) (incl. Auto Restart & Integrated UPS Control)	B7999PS (incl. 12.5 kVA UPS & 40 A hardware kit) NOTE: REQUIRES one of the A1 Panels shown at left.

Table 9-6 LightSpeed Partial UPS Back-up Options

Section 6.0

Contractor (Customer) Supplied Wiring

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length									
3		POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3		POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2- 1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							

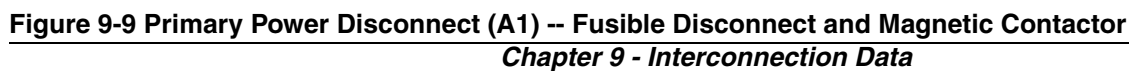
Table 9-7 Runs 1, 2, 3, 4 and 5 Connections

6.1 Primary Power Disconnect

Note: A LOTO-compatible disconnect is required to install this LightSpeedsystem. If a UPS is required, a GE Disconnect is strongly recommended for safe operation. The GE Disconnect and UPS are designed to work together.



Figure 9-8 Primary Power Disconnect (A1)



6.2 Scan Room Warning Light & Door Interlock

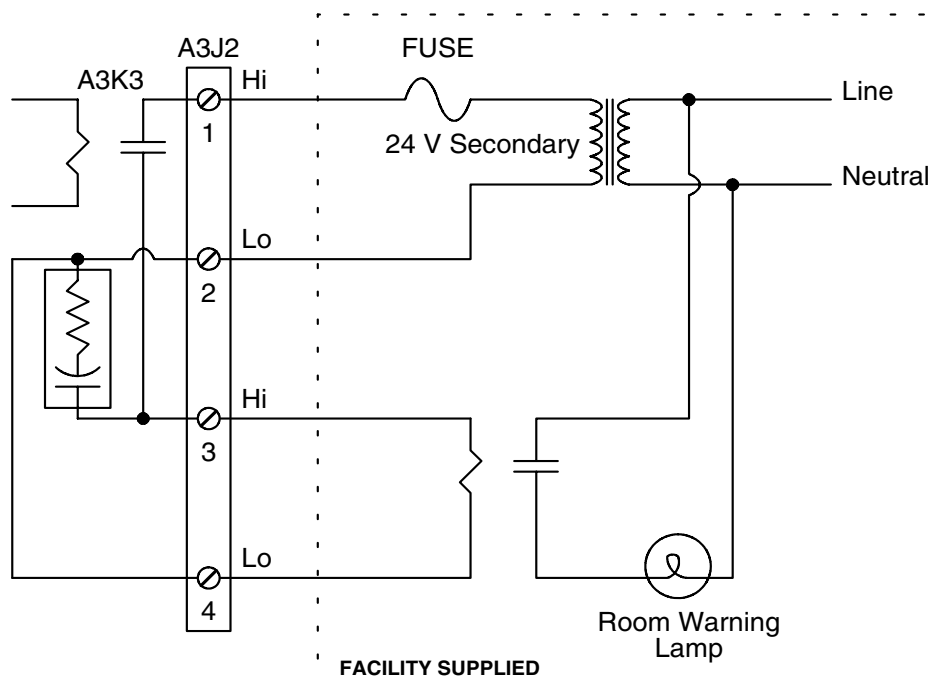


Figure 9-10 Typical Customer Supplied Scan Room – Warning Light Connection to CPDU

Figure 9-11 Typical Customer Supplied Scan Room – Warning Light Connection to NGPDU

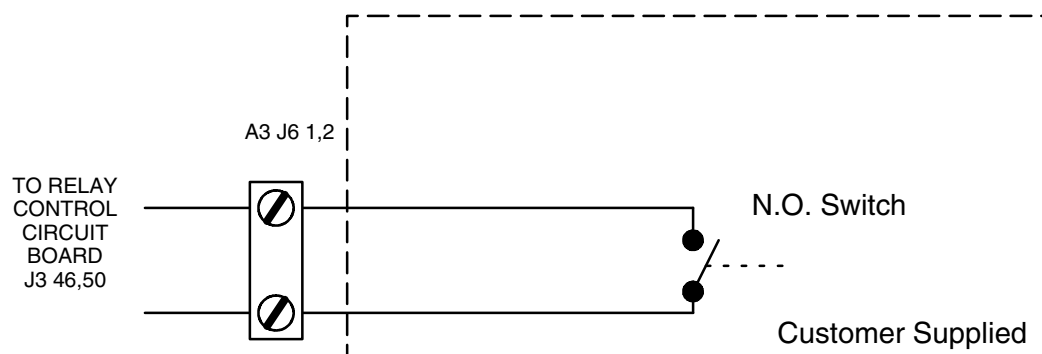


Figure 9-12 Typical Door Interlock Connection to CPDU

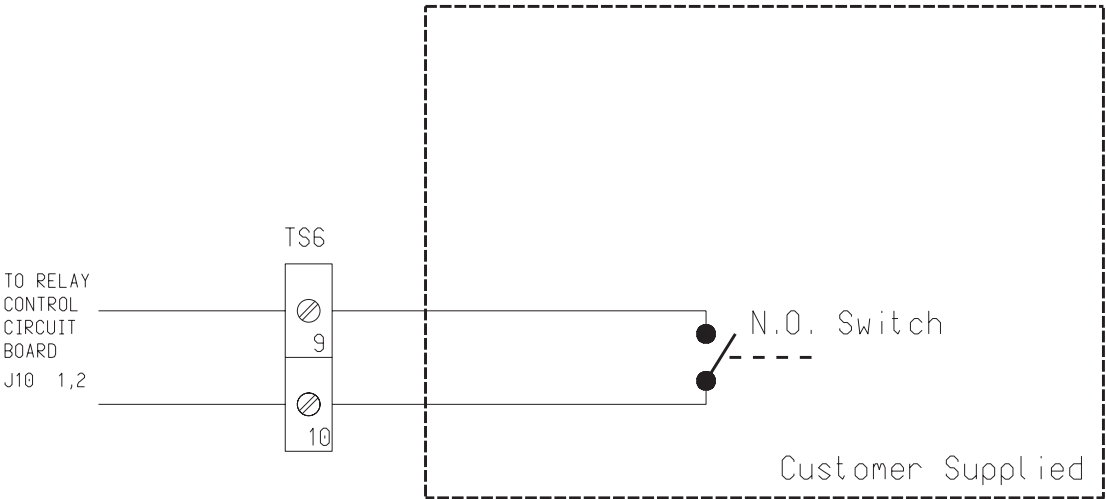


Figure 9-13 Typical Door Interlock Connection to NGPDU

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